From Possibility to Revision - A Candidate Five-Regime Control Architecture for Persistent, Self-Revising Worlds

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Abstract

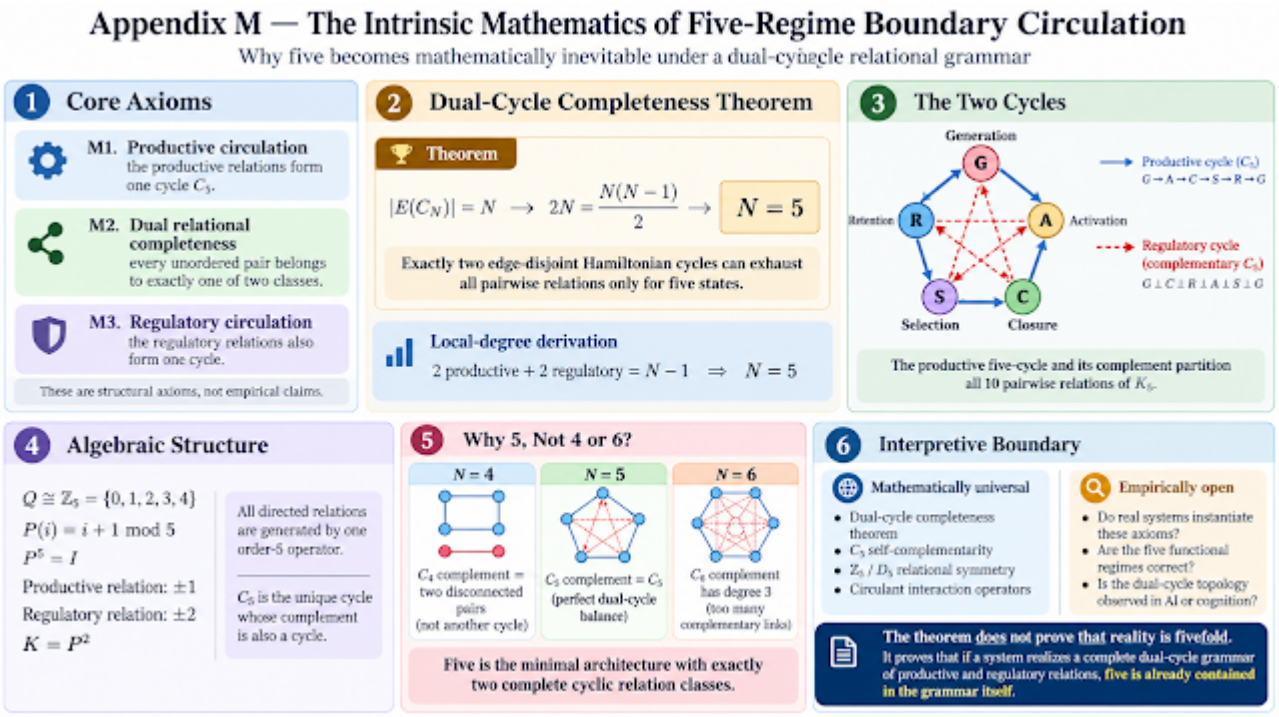
Intelligent systems are usually modeled as state-update processes: given a world model, an agent observes, predicts, acts, learns, and updates its internal state. This picture becomes incomplete once the world model itself can fail. A sufficiently persistent and bounded intelligence must then solve a harder problem: how to generate alternative representations, expose them to consequences, decide what counts as its current effective world, discriminate what should survive, preserve both successful history and unresolved anomalies, and eventually revise the world-model without destroying continuity of identity.

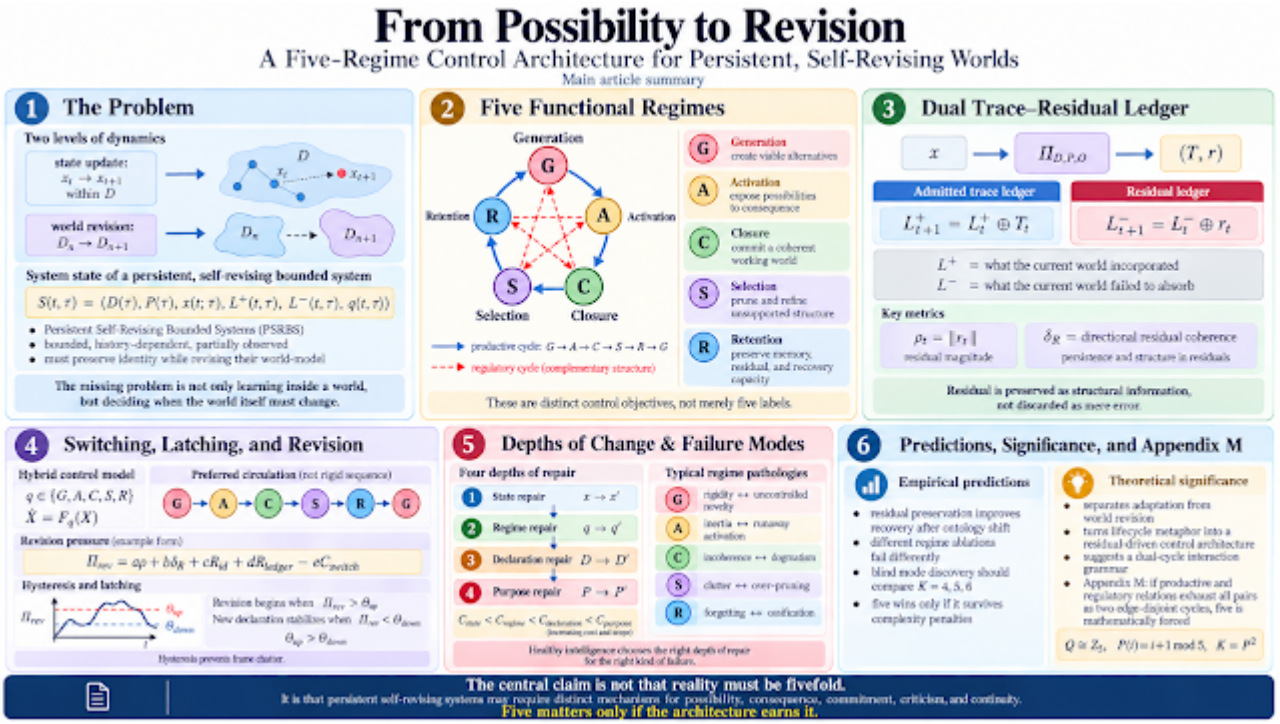
This article develops a candidate control architecture for that problem. Starting from the boundary-formation framework of Semantic Meme Field Theory (SMFT), but without assuming any traditional fivefold classification, we ask how many dynamically distinct functional regimes are minimally required by a persistent self-revising bounded system. A provisional decomposition emerges into five functions: Generation, Activation, Closure, Selection, and Retention. The proposal is not that every intelligent system must contain five physically separate modules, nor that these functions must execute as a rigid pipeline. The stronger and testable hypothesis is that these five functions may form a useful minimal coarse-graining of the control dynamics required for world formation, world use, world evaluation, historical consolidation, and world revision.

The framework distinguishes operational state change from declaration change, introduces a dual trace–residual ledger, treats residual as preserved structural information rather than disposable error, and represents regime transitions through a hysteretic hybrid dynamical system. Competing four- and six-regime models are retained explicitly as null hypotheses. The theory therefore makes the number of regimes an empirical question rather than a premise.

A secondary historical comparison with the Chinese Wuxing, or Five Phases, will be introduced only after the engineering architecture has been derived independently. The primary claim of the article is consequently not cultural but structural: persistent self-revising systems may require separate mechanisms for possibility generation, causal enactment, world commitment, critical discrimination, and continuity through memory and residual preservation.

Keywords: Semantic Meme Field Theory, boundary formation, world-model revision, hybrid control, residual, declaration, adaptive systems, metastability, self-revising agents, persistent intelligence





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1. The Missing Problem in Intelligent Systems

A large fraction of modern intelligence theory begins after a decisive assumption has already been made.

A world has been specified.

Within that world there are states, observations, actions, variables, hypotheses, rewards, predictions, or beliefs. The intelligent system then performs some combination of inference, control, optimization, learning, planning, and memory.

In abstract form:

$x_{t+1} = F_D(x_t,u_t).$ (1.1)

Here D denotes the effective declaration of the world: which variables exist, which distinctions matter, which relations are admissible, and what counts as a valid observation.

Equation (1.1) can describe remarkably sophisticated behavior. Yet it assumes that D remains sufficiently adequate.

The more difficult case begins when the failure lies not in x but in D.

A system can then improve its state indefinitely while remaining trapped inside an inadequate description of reality.

This suggests two fundamentally different kinds of change:

$x_t \rightarrow x_{t+1}$ within the current world, (1.2)

and

$D_n \rightarrow D_{n+1}$ change of the effective world itself. (1.3)

The distinction is central to the later development of the SMFT boundary-formation program. The existing framework explicitly separates ordinary operational evolution from changes in the declaration or world-frame, including distinct operational and revision timescales.
[探討 SMFT 串聯其它名門正派成為五行流轉的可行性]

This article takes the second problem as primary.

> How can a bounded system form an effective world, live inside it, discover what the world cannot absorb, preserve that failure as information, and revise the world without simply destroying itself?

That problem is broader than ordinary learning.

It is the problem of **persistent world revision**.

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2. Persistent Self-Revising Bounded Systems

To avoid making claims about every possible dynamical system, we define a restricted target class.

A **Persistent Self-Revising Bounded System**, abbreviated PSRBS, is a system satisfying at least the following properties:

1. It possesses finite or bounded observational and computational capacity.
2. It persists through time rather than solving only a single isolated task.
3. It acts under partial observation.
4. Its future behavior depends on accumulated history.
5. It can modify internal states within an existing model.
6. It can, at least in principle, modify the model or declaration itself.
7. It must preserve some continuity of identity across revision.

These conditions already imply that ordinary error minimization is insufficient.

If every discrepancy is treated only as a signal to update parameters inside the current model,

the system cannot distinguish:

“my current state estimate is wrong”

from

“the ontology under which I am estimating states is wrong.”

That distinction is essential.

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3. A Minimal State Description

Let the system be represented by:

$S(t,\tau) = (D(\tau), P(\tau), x(t;\tau), L^{+}(t,\tau), L^{-}(t,\tau), q(t,\tau)).$ (3.1)

The components are:

D = current declaration or effective world
P = persistent purpose or reference structure
x = operational state within D
L^{+} = admitted trace ledger
L^{-} = residual ledger
q = dominant control regime

Two time variables are retained:

t = operational time within a current declaration, (3.2)

τ = declaration or revision time across world-frames. (3.3)

The distinction need not imply two physically fundamental clocks. It is initially a modeling separation between two qualitatively different scales of change.

The current world evolves through:

$\dot{x} = F_{D,P}(x,u).$ (3.4)

The declaration itself evolves more slowly:

$dD/d\tau = U_D(D,P,L^{+},L^{-}).$ (3.5)

Purpose may evolve on an even slower scale:

$dP/d\tau_P = U_P(P,L^{+},L^{-}).$ (3.6)

A plausible timescale hierarchy is therefore:

$\tau_x \ll \tau_D \ll \tau_P.$ (3.7)

This hierarchy is a hypothesis, not a universal law. It expresses the intuitive requirement that identity-defining reference structures should normally change less frequently than ordinary states.

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4. Why One Objective Is Not Enough

A conventional optimizer often assumes that a single objective J can coordinate the whole system:

$\dot{S} = -\nabla J(S)$. (4.1)

For persistent self-revising systems this can be problematic.

Different requirements may conflict.

A system sometimes needs more alternatives.

At other times it needs fewer.

Sometimes it should amplify a possibility.

At other times it should suppress it.

Sometimes it should preserve stability.

At other times stability itself has become the problem.

This means that “good behavior” may not correspond to descending one fixed scalar landscape.

Instead the system may require multiple control objectives:

$J_q(S), \quad q \in Q$. (4.2)

The active objective depends on the current regime.

The resulting system is naturally represented as a hybrid dynamical system:

$\dot{S} = F_q(S), \quad q \in Q$. (4.3)

and regime transitions satisfy:

$q_i \to q_j$ iff $g_{ij}(S) \geq 0$. (4.4)

The central theoretical question becomes:

> What is the smallest useful set Q?

No answer should be inserted in advance.

Three-, four-, five-, six-, and higher-regime descriptions are all logically possible.

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5. Deriving Functional Requirements Before Counting Regimes

Instead of starting from a number, begin with failure prevention.

A persistent self-revising system must somehow solve at least five qualitatively different problems.

Problem A — Novelty

If the current representation is inadequate, the system must create alternatives.

Problem B — Consequence

Alternatives must sometimes become operational so that their consequences can be observed.

Problem C — Constitution

The system must decide which relations can jointly constitute its current working world.

Problem D — Criticism

Committed structure must be evaluated, pruned, repaired, or rejected.

Problem E — Continuity

Useful history and unresolved anomalies must survive long enough to influence future revision.

These requirements yield five provisional functions:

G = Generation / Differentiation
A = Activation / Enactment
C = Closure / Integration / Commitment
S = Selection / Refinement
R = Retention / Latency / Recovery

Thus:

$Q_5 = \{G,A,C,S,R\}.$ (5.1)

The notation does **not** yet mean that five is proven minimal.

It means that five functions have survived the first attempt at decomposition.

The canonical candidate circulation is:

$G \rightarrow A \rightarrow C \rightarrow S \rightarrow R \rightarrow G.$ (5.2)

This should not be interpreted as a mandatory sequence.

Its status is initially:

a candidate low-conflict circulation path.

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6. Regime G — Generation and Differentiation

The first problem is straightforward.

A system that can only optimize within its current representational space eventually becomes trapped when the environment changes beyond that space.

It needs a mechanism capable of creating:

- new hypotheses;
- new variables;
- new abstractions;
- new boundaries;

- new candidate declarations;
- new decompositions of previously unresolved structure.

Call the corresponding regime G.

Let V denote viable candidate variety.

A Generation-dominant regime should generally satisfy:

$dV/dt > 0.$ (6.1)

The word “viable” is essential.

Pure random variation can increase nominal diversity without increasing useful possibility.

Generation should therefore be constrained by existing history and purpose.

A schematic objective is:

$J_G = -V + \lambda_G C_{hist} + \mu_G C_{purp}.$ (6.2)

where:

C_{hist} = inconsistency with historical constraint
C_{purp} = inconsistency with persistent purpose

Healthy Generation therefore means:

increase meaningful possibility without severing continuity.

6.1 Under-generation

If G is too weak, the system becomes incapable of escaping the current declaration.

Typical symptoms include:

- conceptual rigidity;
- failure under distribution shift;
- repetitive reasoning;
- inability to invent new latent variables;
- persistent fitting of anomalies into an obsolete ontology.

This can be called **representational sterility**.

6.2 Over-generation

If G dominates excessively:

- hypothesis count explodes;
- boundaries become unstable;
- commitments are repeatedly abandoned;
- the system spends resources producing possibilities it never tests.

The pathology is not “too much intelligence.”

It is uncontrolled expansion of possibility.

A bounded system therefore cannot remain permanently generative.

It requires transition into another regime.

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7. Regime A — Activation and Enactment

Generating a representation does not establish its usefulness.

The system must sometimes expose a candidate to causal consequence.

This includes:

- action;
- inference under the candidate model;
- intervention;
- propagation;
- simulation;
- exploration;
- environmental testing.

Call this regime A.

Let E denote effective causal exposure.

Then:

$dE/dt > 0$ during dominant Activation. (7.1)

A schematic control objective is:

$J_A = -E + \lambda_A B_A + \mu_A I_A.$ (7.2)

where:

B_A = energetic or computational burden
I_A = instability or runaway-amplification cost

The distinction between generation and enactment is critical.

A candidate representation can exist without being allowed to alter the world.

This separation provides a form of sandboxing.

7.1 Why G and A should not be merged casually

Suppose every sufficiently salient generated hypothesis is automatically enacted.

Then:

Generation ≈ Action. (7.3)

The system loses the difference between:

“this possibility is worth representing”

and

“this possibility is safe or useful to execute.”

This produces an **Ideation–Action Collapse**.

The likely consequences include:

- unsafe exploration;
- intervention before adequate evaluation;
- inability to maintain counterfactual possibilities as merely counterfactual.

This supplies the first serious reason to keep G and A distinct.

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8. Regime C — Closure, Integration, and Commitment

Activation alone still does not create a world.

A candidate may temporarily work without deserving incorporation into the persistent structure of the system.

The next problem is constitutional:

> Which relations are allowed to count together as the current operational world?

This is the role of C.

Closure performs a transformation of the general form:

Possibility → Admitted World + Residual. (8.1)

The distinction between admitted content and residual is already central to the mature SMFT boundary-formation program, in which a world is not merely observed but declared, gated, traced, ledgered, and later potentially revised.
[探討 SMFT 串聯其它名門正派成為五行流轉的可行性]

A generic declaration operation can be written:

$(D', r) = \mathcal{C}(X, P, L^{+})$. (8.2)

where r represents what the attempted closure failed to assimilate.

8.1 Closure objective

Let:

Coh(D) = internal coherence
Comp(D,P,L^{+}) = compatibility with purpose and admitted history
Bnd(D) = boundary complexity

Then:

$J_C(D) = -\mathrm{Coh}(D) - \alpha\, \mathrm{Comp}(D,P,L^{+}) + \beta\, \mathrm{Bnd}(D)$. (8.3)

A declaration may be considered latched when:

$J_C(D) \leq \theta_C \quad \text{for } \Delta\tau \geq T_C. \quad (8.4)$

The precise functional form is not yet fundamental.

The important point is structural:

Closure is the regime that converts provisional relations into accountable world structure.

8.2 Closure is not merely stability

A stable attractor need not constitute an effective world.

A world requires more than persistence.

It requires:

- boundary;
- admissibility;
- relation structure;
- interpretability to the observer;
- compatibility with retained history;
- sufficient internal closure for operations to be evaluated.

Thus:

$\text{Persistence} \neq \text{Declaration}. \quad (8.5)$

This distinction prevents the theory from reducing “world formation” to ordinary attractor dynamics.

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9. Regime S — Selection and Refinement

Once a world has been committed strongly enough to produce accountable consequences, it must be criticized.

Selection asks:

> What should survive?

Its operations include:

- measurement;
- falsification;
- comparison;
- pruning;
- compression;
- repair;
- residual attribution.

Let K denote active structural complexity.

A typical Selection tendency is:

$dK/dt < 0$ subject to preserved function. (9.1)

A schematic objective is:

$J_S = K + \lambda_S \text{Loss}(F) + \mu_S \text{Loss}(I)$. (9.2)

where:

F = operational fitness
I = identity preservation

This means that S should not simply minimize complexity.

It should minimize unsupported complexity.

9.1 Why Closure and Selection are different

This distinction is one of the most important in the article.

Closure asks:

> What constitutes the current admissible world?

Selection asks:

> Which parts of that world, or candidate additions to it, survive evidence and evaluation?

If these two functions are merged, the same mechanism determines both:

1. what counts as admissible evidence;
2. whether that evidence validates the frame.

That creates a risk of circularity.

The system can become a court that writes its own constitution and judges all cases under rules that cannot themselves be questioned.

Thus:

$C \neq S$ is presently a strong hypothesis. (9.3)

The principal four-regime null model will challenge exactly this distinction.

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10. Regime R — Retention, Latency, and Recovery

Selection cannot simply delete everything that is not currently useful.

A persistent self-revising system needs to preserve at least:

- successful traces;
- identity-bearing history;
- dormant but recoverable alternatives;

- unresolved anomalies;
- resources and capacity for future cycles.

This is regime R.

Let:

M = recoverable memory
A_e = active excitation
Oss = ossification cost

Then:

$J_R = -M + \lambda_R A_e + \mu_R \mathrm{Oss}.$ (10.1)

Healthy Retention satisfies approximately:

$dA_e/dt < 0, \quad dM/dt \approx 0.$ (10.2)

The goal is not maximal freezing.

It is stable recoverability.

Existing Proto-Eight work already treats retention as an active dynamic involving memory wells, decay, resurfacing, and the danger of over-locking rather than as passive storage. Proto-Eight Collapse Geometry_S…

Likewise, its coupled “Ventilate–Store” construction explicitly describes a breathing process of intake, coarse-graining, deposition, and later resurfacing. Proto-Eight Collapse Geometry_S…

10.1 Why Selection and Retention are different

Suppose every rejected structure is erased and every accepted structure is stored.

Then:

Selection ≈ Memory. (10.3)

This is dangerous because the present world decides what deserves to remain available to challenge the present world later.

A system can become permanently self-sealing.

What is currently rejected may later become crucial after an environmental shift.

Therefore a robust architecture needs the ability to preserve:

“not admitted now”

without converting it into:

“never existed.”

This observation will motivate the residual ledger in the next part of the article.

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11. First Result: Five Distinct Control Objectives

The decomposition now gives five different dominant tendencies:

G — expand structured possibility
A — expand causal consequence
C — increase coherent commitment
S — reduce unsupported structure
R — reduce active excitation while preserving recoverability

These are not five synonyms.

They optimize different quantities.

A system can therefore exhibit:

high G with low A,

high A with weak C,

strong C with weak S,

strong S with poor R,

or strong R with suppressed G.

This leads naturally to distinct failure modes.

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12. Characteristic Failure Signatures

Regime	Deficiency	Excess
Generation	rigidity, no alternatives	fragmentation, uncontrolled novelty
Activation	inertia, untested models	runaway propagation, exhaustion
Closure	incoherence, inability to commit	dogmatism, premature lock-in
Selection	error accumulation, clutter	brittleness, destructive pruning
Retention	forgetting, identity loss	ossification, inability to reawaken

A serious regime theory should predict approximately separable failure signatures.

If targeted interventions on the five functions all collapse onto one underlying failure dimension, then the fivefold decomposition is probably descriptive rather than structural.

That gives the theory a straightforward route to falsification.

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13. The Candidate Five-Regime Hypothesis

We can now state the first formal hypothesis.

> **Five-Regime Functional Hypothesis.**
> A persistent self-revising bounded system requires mechanisms functionally equivalent to Generation, Activation, Closure, Selection, and Retention if it is to preserve both adaptability and continuity under possible world-model failure.

This is weaker than saying that the five functions must correspond to five neural modules.

It is also weaker than saying that the system must cycle through them in a fixed order.

A stronger hypothesis is:

> **Five-Regime Modal Hypothesis.**
> Under appropriate coarse-graining, the five functions tend to appear as dynamically distinguishable metastable control regimes.

Stronger still:

> **Five-Regime Minimality Hypothesis.**
> Five is the smallest coarse-grained regime count that preserves all five functional distinctions without destructive merger.

These three claims must be tested separately.

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14. The Central Question for the Next Sections

At this point the architecture exists, but the most SMFT-specific mechanism has not yet been introduced.

A conventional lifecycle model could still say:

Generation → Activation → Closure → Selection → Retention.

But that would leave unanswered:

> **What makes the system move?**

Why should Retention return to Generation?

Why should Activation end?

Why should Closure fail?

What makes a current world become revisable?

The next sections therefore introduce the proposed circulation driver:

Residual

and, more specifically:

> **a dual historical structure in which admitted traces and unresolved residuals are both preserved.**

That is where the five-regime lifecycle begins to turn into an SMFT-style theory of self-revising worlds.

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15. The Dual Trace–Residual Ledger

The five-regime cycle is not yet a theory of self-revision unless the system can preserve two very

different kinds of history.

Most adaptive architectures preserve what was accepted:

- successful actions;
- useful memories;
- learned parameters;
- validated hypotheses;
- compressed summaries.

That produces a conventional history of admitted structure.

SMFT suggests a second history is equally important:

> what repeatedly failed to fit the current declaration.

The mature boundary-formation framework explicitly treats residual as information that should be preserved, ledgered, and allowed to influence later revision rather than silently discarded as mere error.
[探討 SMFT 串聯其它名門正派成為五行流轉的可行性]

This motivates a dual ledger.

Let the observer–declaration interface produce:

$(T_t, r_t) = \Pi D,P,O(x_t).$ (15.1)

where:

T_t = admitted trace
r_t = residual

Then update:

$L^+_{t+1} = L^+_t \oplus T_t.$ (15.2)

$L^-_{t+1} = L^-_t \oplus r_t.$ (15.3)

The system therefore accumulates two histories:

L^+ = history of what became part of the effective world

L^- = history of what the effective world repeatedly failed to absorb.

This distinction is not a bookkeeping detail. It is one of the mechanisms that makes genuine world revision possible.

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16. Why Ordinary Error Minimization Is Not Enough

Suppose a system encounters a discrepancy e_t.

A conventional adaptive strategy may update parameters to minimize:

$E = \Sigma_t \|e_t\|^2.$ (16.1)

That is entirely appropriate when the model class is adequate.

The problem begins when the discrepancy arises because the model class itself is inadequate.

There are then at least two qualitatively different explanations:

H_1: the current world is correct, but its state estimate is poor;

H_2: the current world is structurally inadequate.

Ordinary error reduction tends to treat both as H_1.

The system attempts:

$\theta \to \theta'$ inside D. (16.2)

SMFT requires the additional possibility:

$D \to D'$. (16.3)

Residual is what allows evidence for H_2 to survive long enough to become visible.

Without a residual ledger, the model can continually absorb, smooth, reinterpret, or forget anomalies.

The result is **ontology lock-in**.

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17. Residual Must Be More Than a Scalar

A single error magnitude cannot determine whether the declaration is wrong.

A large residual can arise from:

- noise;
- transient disturbance;
- adversarial corruption;
- insufficient computation;
- local state error;
- genuine structural mismatch.

Conversely, a small residual may be highly important if it is persistent and directionally coherent.

Therefore write residual as a structured object:

r_t = (rfit, rdir, rid, robs, rcost, rmem, ...). (17.1)

Possible components include:

rfit = local mismatch under the current model

rdir = direction of systematic model inadequacy

rid = identity or purpose inconsistency

robs = observer incompatibility

rcost = resource or viability pressure

rmem = contradiction with historical ledger

The precise decomposition is application-dependent.

The general principle is:

> residual should preserve information about the *form* of failure, not merely its magnitude.

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18. Residual Magnitude and Directional Persistence

Define total residual magnitude:

$\rho_t = \|r_t\|$. (18.1)

Magnitude tells us how badly something currently fits.

But now define directional coherence over a window W:

$\delta R = \|\Sigma_{k \in W} w_k\, r_k\| \,/\, \Sigma_{k \in W} w_k \|r_k\|$. (18.2)

Then:

$0 \le \delta R \le 1$. (18.3)

Interpretation:

$\delta R \approx 0$
means residual directions cancel.

$\delta R \approx 1$
means residual repeatedly points toward a similar structural correction.

The earlier SMFT development already distinguishes residual magnitude from directional residual: a large undirected residual can be noise, whereas a smaller but consistently aligned residual may indicate structural bias. $\mathbb{O} \to G_2_SO(4) \to \mathbb{H} \to \mathbb{C}^2$ 成界過程初探 1…

This is one of the most important mechanisms available for deciding whether a system should adapt *within* the current world or revise the world itself.

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19. State Error Versus Declaration Error

This distinction can now be formalized.

Let:

ρlocal = residual explainable by state or parameter adjustment

ρstruct = residual remaining after admissible local adaptation

Then:

ρtotal = ρlocal + ρstruct schematically. (19.1)

A local update is justified when:

ρlocal ≫ ρstruct. (19.2)

A declaration revision becomes plausible when:

ρstruct persists and δR remains high. (19.3)

Thus the system should not ask only:

“How large is the error?”

It should ask:

> “Can the current world absorb this error without changing what counts as the world?”

That is a much deeper diagnostic question.

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20. Revision Pressure

We can now define a generic revision pressure.

Let:

Πrev = aρstruct + bδR + cRid + dRledger − eCswitch. (20.1)

where:

Rid = identity inconsistency

Rledger = contradiction accumulated in historical records

Cswitch = cost of abandoning the current declaration

and a, b, c, d, e are positive weights.

A declaration revision becomes favourable when:

Πrev > Θrev. (20.2)

This equation is only schematic.

Its conceptual importance is greater than its specific form.

It transforms “phase transition” from an analogy into an engineering question:

> What measurable quantities accumulate until continuation inside the present world becomes more costly than revision?

---

21. Latching

A world should not be revised whenever Πrev briefly crosses one noisy threshold.

Persistent systems need **latching**.

A declaration D remains active even under moderate perturbation.

Formally, use two thresholds:

$\Theta_{up} > \Theta_{down}$. (21.1)

Revision begins when:

$\Pi_{rev} > \Theta_{up}$. (21.2)

A new declaration becomes stably latched only after:

$\Pi_{rev} < \Theta_{down}$. (21.3)

This creates hysteresis.

The current world therefore possesses a basin of persistence rather than a knife-edge boundary.

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22. Why Hysteresis Is Necessary

Without hysteresis, a system near a boundary can oscillate rapidly:

$D_1 \to D_2 \to D_1 \to D_2 \to \dots$ (22.1)

This is **frame chatter**.

Frame chatter is dangerous because changing a world-model is not free.

It may require:

- reinterpretation of memory;
- reassignment of meaning;
- recomputation of plans;
- re-evaluation of identity;
- rebuilding of interfaces;
- discarding incompatible internal structure.

Therefore declaration changes should usually exhibit inertia.

Hysteresis supplies that inertia.

The same logic applies to regime changes.

For each transition $q_i \to q_j$, define distinct entry and exit conditions.

A regime should be easy enough to enter when necessary but difficult enough to leave that it can complete useful work.

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23. The Five Regimes as a Hybrid Automaton

We can now formalize the candidate control architecture.

Continuous state:

$X = (x, D, P, L^+, L^-)$. (23.1)

Discrete regime:

$q \in Q = \{G,A,C,S,R\}$. (23.2)

Within each regime:

$\dot{X} = Fq(X)$. (23.3)

A transition occurs when:

$q_i \to q_j$ iff $g_{ij}(X) \geq 0$. (23.4)

If the transition changes latent structure, apply a reset map:

$X^+ = Rij(X^-)$. (23.5)

This hybrid-system representation immediately clarifies several issues.

The five regimes are not substances.

They are not permanent components.

They are not necessarily executed in strict order.

They are **dynamical operating modes**.

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24. The Canonical Circulation

The proposed low-conflict circulation is:

$G \to A \to C \to S \to R \to G$. (24.1)

The interpretation is:

Generation produces viable alternatives.

Activation exposes selected alternatives to consequence.

Closure converts sufficiently coherent consequences into an admitted working world.

Selection evaluates and refines that world.

Retention consolidates what should persist while preserving unresolved remainder.

Residual-informed history then biases the next round of Generation.

The key phrase is:

low-conflict circulation.

This is not claimed to be the only legal path.

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25. Generation → Activation

A candidate should move from G to A only if it has enough structure to justify enactment.

Define candidate viability:

$V(c) = V_{struct}(c) + V_{purpose}(c) + V_{history}(c).$ (25.1)

Then:

$G \rightarrow A$ if $V(c) > \theta_A$. (25.2)

This separates:

representation

from

intervention.

That separation is necessary for safe exploration.

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26. Activation → Closure

A candidate may be experimentally useful without deserving incorporation into the current world.

Closure becomes relevant when enacted structure exhibits sustained coherence and consequence.

For example:

$A \rightarrow C$ if $Coh(c) > \theta_{coh}$ and $Cons(c) > \theta_{cons}$ for $\Delta t > T_{coh}$. (26.1)

where:

Coh = structural coherence

Cons = consequentiality

The system is then forced to answer:

> Is this still an experiment, or has it become part of the world I rely upon?

That is a declaration problem.

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27. Closure → Selection

Once a declaration becomes sufficiently latched, evaluation becomes meaningful relative to a stable reference.

Thus:

$C \rightarrow S$ if $Latch(D)=1$ and $Evidence(D)>\theta_{eval}$. (27.1)

This ordering has a subtle advantage.

A system cannot perform strong comparative judgment unless it possesses some temporary standard against which judgments are made.

Closure supplies that standard.

Selection subsequently challenges it.

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28. Selection → Retention

Selection need not eliminate every unresolved discrepancy.

It needs to determine which discrepancies require immediate action and which can be preserved.

Thus:

$$S \to R \quad \text{if} \quad \text{ActiveCorrectionLoad} < \theta R. \quad (28.1)$$

and:

$$\text{CriticalResiduals} \in \{\text{resolved, deferred, preserved}\}. \quad (28.2)$$

The word “preserved” matters.

A system does not need to solve everything before entering Retention.

It needs to know what remains unresolved.

---

29. Retention → Generation

This is arguably the most important transition in the entire architecture.

A healthy system should not restart Generation randomly.

Generation should be biased by unresolved history.

Define:

$$G_{n+1} \sim \mathcal{P}(c \mid P, L^{+}, L^{-}). \quad (29.1)$$

Then new candidate formation depends jointly on:

what succeeded

and

what remained unexplained.

A simple trigger is:

$$R \to G \quad \text{if} \quad \rho\text{struct} > \rho\text{min} \quad \text{and} \quad \delta R > \delta\text{min}. \quad (29.2)$$

This gives a concrete interpretation to:

> failure becomes the seed of novelty.

The next world is generated partly from what the previous world could not contain.

---

30. Two Coupled Cycles

The architecture now reveals two different but coupled loops.

Structural loop

$D_n \to x(t|D_n) \to \text{Observation} \to \text{Trace/Residual} \to \text{Ledger} \to \text{Revision} \to D_{n+1}.$ (30.1)

Regime loop

$G \to A \to C \to S \to R \to G.$ (30.2)

The structural loop describes **what happens to the world-model**.

The regime loop describes **what dominant control objective governs the process**.

These should not be conflated.

A system can be in Selection without immediately changing D.

Likewise, Generation can occur locally without world revision.

True declaration change occurs only when residual and revision pressure cross the appropriate threshold.

---

31. Three Depths of Change

This suggests a useful hierarchy.

Level 1 — State change

$x \to x'.$ (31.1)

The world remains unchanged.

Level 2 — Regime change

$q \to q'.$ (31.2)

The dominant control objective changes.

Level 3 — Declaration change

$D \to D'.$ (31.3)

The effective world itself changes.

A still deeper level may exist:

Level 4 — Purpose change

$P \to P'.$ (31.4)

This modifies what counts as meaningful continuity.

These four kinds of change should not be treated as equivalent.

---

32. Why Purpose Must Change More Slowly

If purpose or identity constraints change as quickly as ordinary states, continuity becomes ill-defined.

Therefore one expects approximately:

$\tau_x \ll \tau_q \leq \tau_D \ll \tau_P.$ (32.1)

This ordering need not hold universally.

But it expresses a design principle:

> the deeper the revision, the greater the evidential burden and switching cost.

A stable architecture should normally prefer:

state repair before regime repair,

regime repair before declaration repair,

declaration repair before purpose repair.

---

33. Revision Escalation

A possible escalation policy is:

Level 1 repair if $\rho_{struct} < \theta_1$. (33.1)

Level 2 repair if $\theta_1 \leq \rho_{struct} < \theta_2$. (33.2)

Level 3 repair if $\theta_2 \leq \rho_{struct} < \theta_3$. (33.3)

Level 4 repair if $\rho_{struct} \geq \theta_3$ and identity contradiction persists. (33.4)

This provides an engineering interpretation of “deep self-revision.”

The system does not rewrite itself simply because ordinary adaptation becomes inconvenient.

It escalates only when shallower repair repeatedly fails.

---

34. Identity Across World Revision

World revision raises a difficult question.

If D changes substantially, why call the resulting process the same system?

Continuity must be carried by something deeper than the current world representation.

Let:

$I = I(P, L^+, L^-, D).$ (34.1)

be an identity-continuity functional.

A revision is admissible only if:

$I(D_{n+1},P,L^{+},L^{-}) \geq I_{min}.$ (34.2)

This does not imply that identity is immutable.

It means that revision must preserve a traceable chain of accountability.

The ledger is therefore not merely memory.

It is part of what makes self-revision distinguishable from replacement.

---

35. The 2+1+2 Structure

The five-regime architecture displays a striking asymmetry.

Generation and Activation are broadly expansive.

Selection and Retention are broadly contractive.

Closure lies between them.

So:

$(G,A) \mid C \mid (S,R).$ (35.1)

This suggests a deeper candidate structure:

two outward operations
one central world-forming operation
two inward operations.

---

36. Outward Operation I — Expand Possibility

Generation increases the set of accessible candidate structures.

Let ΩP denote accessible possibility volume.

Then:

$d\Omega P/dt > 0$ under G. (36.1)

This is expansion in representational possibility.

---

37. Outward Operation II — Expand Consequence

Activation increases the causal footprint of selected possibilities.

Let ΩE denote enacted influence.

Then:

$d\Omega E/dt > 0$ under A. (37.1)

This is expansion in realized consequence.

Thus the outward pair performs:

potential expansion → realized expansion. (37.2)

---

38. Central Operation — Closure

Closure is qualitatively different.

Its primary action is neither simple expansion nor contraction.

It changes **admissibility topology**.

A diffuse set of partially compatible relations becomes a bounded operational whole.

Schematically:

many candidates → one declared coherence class. (38.1)

Closure therefore acts as a bridge between proliferation and accountability.

---

39. Inward Operation I — Contract Unsupported Structure

Selection reduces unsupported complexity.

Let ΩK denote active representational complexity.

Then:

$d\Omega K/dt < 0$ under S. (39.1)

This is contraction by discrimination.

---

40. Inward Operation II — Contract Active Excitation

Retention reduces active processing while preserving recoverability.

Let ΩE^* denote active excitation.

Then:

$d\Omega E^*/dt < 0$ under R. (40.1)

This is contraction into latency.

Thus the inward pair performs:

structural contraction → energetic/operational contraction. (40.2)

The entire architecture becomes:

expand possibility → expand consequence → close world → contract structure → contract activity → reopen possibility. (40.3)

This is the strongest structural reason yet for taking five seriously.

---

41. The 2+1+2 Boundary Control Conjecture

We can state a secondary hypothesis:

> **2+1+2 Boundary Control Conjecture.**
> Persistent self-revising bounded systems may require two functionally distinct outward transformations, one boundary-forming integration process, and two functionally distinct inward transformations.

In symbolic form:

$\text{Outward}_2 \rightarrow \text{Closure}_1 \rightarrow \text{Inward}_2 \rightarrow \text{Outward}_2$. (41.1)

This is not yet a theorem.

But it is more constrained than an arbitrary five-category taxonomy.

It offers a possible explanation for why five may be a stable coarse-graining rather than a cultural artifact.

---

42. Why Four May Be Too Few

The strongest four-regime competitor is:

$G \rightarrow A \rightarrow CS \rightarrow R$. (42.1)

where Closure and Selection are merged.

This is attractive because many engineering pipelines combine validation and commitment.

However, the merger creates a structural danger.

The same process determines:

- what counts as a legitimate world;
- what counts as valid evidence inside that world.

The result can be self-confirming closure.

Call this **Constitution–Judgment Collapse**.

A system suffering this collapse tends to accept frames that make its own evaluation criteria easy to satisfy.

Its characteristic pathologies should include:

- confirmation bias;
- ontology lock-in;
- reduced sensitivity to frame-challenging anomalies;
- reinterpretation of residual as noise.

These are experimentally testable predictions.

---

43. Other Four-Regime Mergers

G + A

Generation and Activation merge.

Predicted pathology:

counterfactuals leak into action.

A + C

Activation and Closure merge.

Predicted pathology:

successful behavior becomes confused with legitimate world commitment.

S + R

Selection and Retention merge.

Predicted pathology:

rejected hypotheses are forgotten rather than preserved as residual.

R + G

Retention and Generation merge.

Predicted pathology:

memory consolidation and novelty interfere with each other.

Therefore every obvious reduction to four sacrifices an important distinction.

That does not prove five is minimal.

But it gives four a significant burden of proof.

---

44. Why Six May Be Unnecessary

The strongest six-regime alternatives arise by splitting one of the five functions.

For example:

$S \rightarrow S_1 + S_2$

where:

S_1 = evaluation / falsification

S_2 = repair / refinement

or:

$R \rightarrow R_1 + R_2$

where:

R_1 = retention / cooling

R_2 = resurfacing / regeneration.

A split should be accepted only if the two subfunctions support distinct metastable control states.

A useful criterion is:

> A function deserves its own macro-regime only if the system can remain stably dominated by that function without immediately performing its neighbour.

This is stronger than saying the function has a distinct name.

---

45. Selection Versus Refinement

A system can certainly recognize a failure before knowing how to repair it.

Therefore:

evaluation

and

repair

are conceptually distinct.

But they may still belong to one macro-regime if the controller treats both as different operations under the same dominant objective:

reduce unsupported structure while preserving viability.

Thus five can remain the coarser useful description even if lower-level implementation contains six or more substates.

---

46. Retention Versus Regeneration

Likewise, retention and reactivation are distinct events.

But regeneration may be better understood as the transition:

$R \rightarrow G.$ (46.1)

rather than a sixth stable mode.

If so, “Regeneration” belongs to the boundary between cycles rather than to the interior of one

cycle.

This is one reason five currently appears more compact than six.

---

47. Five as a Coarse-Graining Hypothesis

The five-regime architecture should therefore be understood as a coarse-graining.

At finer resolution:

G may contain multiple exploration modes.

A may contain simulation, action, and propagation.

C may contain reconciliation, binding, and latching.

S may contain measurement, falsification, pruning, and repair.

R may contain consolidation, dormancy, resurfacing, and recovery.

The theory does not deny these internal distinctions.

It asks whether they compress naturally into five dominant functional families.

---

48. Model Selection Across K Regimes

The regime count should ultimately be empirical.

Let M_K denote a K-regime model.

Define a model-quality score:

$$Q(K) = \text{PredictiveFit}(K) - \lambda \text{Complexity}(K) - \mu \text{InterventionError}(K). \quad (48.1)$$

Then:

$$K^* = \operatorname{argmax}_K Q(K). \quad (48.2)$$

The five-regime hypothesis predicts:

$$K^* \approx 5 \quad \text{for a significant class of persistent self-revising bounded systems.} \quad (48.3)$$

The phrase “significant class” is deliberate.

The theory does not require all systems to produce K=5.

---

49. Blind Mode Discovery

The strongest experiment should not hard-code G, A, C, S, and R.

Instead let a switching model infer:

$$z_t \in \{1,\ldots,K\}. \quad (49.1)$$

The experimenter should not name the states in advance.

After fitting, test:

1. what K best explains the data;
2. whether inferred states are metastable;
3. whether targeted interventions distinguish them;
4. whether their functional signatures correspond approximately to G, A, C, S, and R.

If:

K = 4

reliably outperforms five, the minimality hypothesis should be weakened.

If:

K = 6

reliably wins, the theory should identify which function split.

If:

K varies by domain,

then five may be one useful mesoscopic resolution rather than a universal architecture.

---

50. Nested Regime Cycles

The five-regime architecture may also be recursive.

At macro scale, a research organization may be in Generation.

Inside it, an experiment can simultaneously be in Selection.

Inside that experiment, a local controller may be in Retention.

Thus:

$$q(\text{micro}) \neq q(\text{meso}) \neq q(\text{macro}). \quad (50.1)$$

The regime label is therefore scale-relative.

This is important.

A statement such as:

“the system is in Generation”

is incomplete unless the observation scale is specified.

---

51. Recursive Five-Regime Structure

Each regime can itself contain a smaller cycle.

Inside Selection, for example:

generate candidate explanations
activate tests
close an evaluation frame
select among explanations
retain the result

So:

$G \to A \to C \to S \to R$

may be recursively embedded inside S.

Likewise inside G, an internal subcycle may determine which candidate-generation strategy to keep.

This recursive possibility fits naturally with the broader SMFT emphasis on self-revising, recursively disclosed worlds.

---

52. Regime Occupancy

Define long-run occupancy:

$\pi q = Tq / Ttotal.$ (52.1)

Then:

$\Sigma q \, \pi q = 1.$ (52.2)

Different environments should produce different occupancy distributions.

A highly stable environment may yield:

$\pi C + \pi R$ high. (52.3)

A rapidly changing environment may yield:

$\pi G + \pi A$ high. (52.4)

A diagnostic or crisis environment may produce:

πS high. (52.5)

This gives the theory a simple empirical handle.

---

53. Regime Imbalance

Persistent imbalance should produce characteristic pathologies.

$\pi G \gg$ others
→ endless exploration.

$\pi A \gg$ others
→ runaway action.

$\pi C \gg$ others

→ excessive institutionalization.

$\pi S \gg$ others
→ chronic criticism and erosion.

$\pi R \gg$ others
→ dormancy and stagnation.

The key idea is not that equal occupancy is healthy.

Healthy occupancy depends on context.

The pathology lies in **failure to transition appropriately**.

---

54. Transition Cost

Switching regimes is not free.

Let:

C_{ij} = cost of transition $q_i \to q_j$. (54.1)

Then a simple controller may choose:

$q_{t+1} = \text{argmin}_j [J_j(S_t) + Cq_t,j + Hj(S_t)]$. (54.2)

where Hj represents hysteresis or commitment penalties.

This prevents a controller from switching whenever another objective becomes marginally better for one instant.

---

55. Preferred Circulation Versus Legal Transition Graph

The pentagonal cycle:

$G \to A \to C \to S \to R \to G$

should be interpreted as a preferred circulation.

The legal graph is richer.

Examples:

$G \to R$
when no viable candidate emerges.

$A \to G$
when testing immediately falsifies the candidate.

$A \to S$
when evidence becomes decisive before closure.

$C \to G$
when integration exposes contradiction.

$S \to A$
when evaluation demands another experiment.

$R \to C$
when accumulated memory can resolve a problem without broad regeneration.

Thus the five-regime model is **not** a deterministic five-step ritual.

This is important both scientifically and rhetorically.

---

56. Three Kinds of System Failure

The architecture now permits a useful distinction.

State failure

x is wrong inside a viable D.

Control-regime failure

q is inappropriate for the current situation.

Declaration failure

D itself is inadequate.

These failures require different remedies.

Treating all three as ordinary prediction error can waste enormous adaptive effort.

---

57. Repair Depth

Correspondingly:

State repair:

$$x \to x'. \quad (57.1)$$

Regime repair:

$$q \to q'. \quad (57.2)$$

Declaration repair:

$$D \to D'. \quad (57.3)$$

Purpose repair:

$$P \to P'. \quad (57.4)$$

This yields a nested repair hierarchy.

A rational system should generally attempt the cheapest adequate repair first.

---

58. A Revision-Cost Principle

Let:

$$C_{state} < C_{regime} < C_{decl} < C_{purpose}. \quad (58.1)$$

Then choose the lowest-depth repair whose expected reduction in residual exceeds its cost.

Formally:

$$k^* = \arg\max_k [\Delta R_k - C_k]. \quad (58.2)$$

where k indexes repair depth.

This turns “self-revision” into a resource-sensitive engineering problem.

---

59. Preventing Gratuitous World Revision

One danger of any theory emphasizing model revision is over-revision.

A system could interpret every surprise as evidence that the world-model must be replaced.

The cost hierarchy prevents this.

World revision should occur only when:

$$\text{ExpectedGain}(D \to D') > C_{decl} + \text{Risk}_{revision}. \quad (59.1)$$

This makes conservative persistence rational.

A stable world is valuable even if imperfect.

---

60. The First Compact Kernel

We can now compress the architecture into four core equations.

Operational flow:

$$\dot{x} = f_q(x; D, P, L^+, L^-). \quad (60.1)$$

Observer split:

$$(T, r) = \Pi_{D,P,O}(x). \quad (60.2)$$

Dual-ledger update:

$$(L^+, L^-) \leftarrow (L^+ \oplus T,\ L^- \oplus r). \quad (60.3)$$

Meta-update:

$$(D, q) \leftarrow U(D, q, P, L^+, L^-). \quad (60.4)$$

with:

$$q \in \{G, A, C, S, R\}. \quad (60.5)$$

This is the smallest useful expression of the current proposal.

It contains:

- world-relative dynamics;
- observation;
- admitted history;
- residual preservation;
- regime switching;
- declaration revision.

---

61. A More Explicit Kernel

A more explicit form is:

$\dot{x} = fq(x,u;D,P).$ (61.1)

$(T_t,r_t) = \Pi D,P,O_t(x_t).$ (61.2)

$L^+_{t+1} = L^+_t \oplus T_t.$ (61.3)

$L^-_{t+1} = L^-_t \oplus r_t.$ (61.4)

$\Pi rev = \mathscr{R}(L^-,D,P,L^+).$ (61.5)

$q_{t+1} = Q(q_t,\Pi rev,L^+,L^-).$ (61.6)

$D_{n+1} = U(D_n,P,L^+,L^-) \quad \text{if} \quad \Pi rev > \Theta up.$ (61.7)

This form is close to something that could be implemented.

---

62. Why This Is More Than a Lifecycle Diagram

A conventional lifecycle says:

explore → act → evaluate → remember.

The present architecture adds at least four deeper features:

1. **world declaration is explicit;**
2. **residual is preserved rather than erased;**
3. **world revision is distinct from state adaptation;**
4. **regime transitions are themselves dynamical and hysteretic.**

These are the features that make the proposal specifically compatible with the mature SMFT boundary-formation program.

---

63. The Strongest Novel Mechanism So Far

The deepest mechanism can be stated simply:

> A world forms by selective admission, and therefore every world necessarily produces a remainder.

That remainder is not merely waste.

If preserved across time, it becomes a structured record of the limitations of the current world.

Therefore:

Declaration → Residual. (63.1)

Residual + Ledger → Revision Pressure. (63.2)

Revision Pressure → Regime Shift / Re-declaration. (63.3)

Re-declaration → New Possibility Structure. (63.4)

This gives self-revision an endogenous source.

The new world arises partly from what the old world excluded.

---

64. A Boundary-Formation Conservation Intuition

A useful heuristic is:

Admitted Structure + Residual ≈ Processed Possibility. (64.1)

This should not yet be treated as a literal conservation law.

But conceptually it prevents a common modeling error:

discarding information simply because the current representation does not contain it.

A future formal theory may define a measure μ such that:

μ(Input) = μ(Admitted) + μ(Residual) + μ(Dissipated). (64.2)

If a disciplined version of this balance can be derived, it could become a powerful part of the SMFT kernel.

For now it remains a research direction.

---

65. The Core Scientific Question

We can now state the central empirical question of the article:

> When a persistent bounded system is required to survive repeated changes in its effective world, does its control dynamics spontaneously separate into approximately five metastable functional regimes corresponding to Generation, Activation, Closure, Selection, and Retention?

This question is testable.

It does not depend on accepting SMFT as a complete theory.

It does not depend on Chinese philosophy.

It does not depend on any particular AI architecture.

And it leaves open the possibility that the answer is four, six, or context-dependent.

---

66. What Would Count as Strong Evidence?

Evidence should require more than visual resemblance.

Strong support would include:

1. blind inference favouring approximately five metastable modes;
2. distinct dwell-time and transition statistics;
3. distinct causal response to targeted interventions;
4. characteristic failure modes under regime-specific ablation;
5. improved adaptation under ontology shift;
6. preservation of frame-challenging residuals;
7. lower unnecessary re-declaration rates than simpler models;
8. cross-domain recurrence of approximately the same functional decomposition.

The strongest result would be independent re-emergence of the five modes across different implementations.

---

67. What Would Falsify the Five-Regime Hypothesis?

Several outcomes would weaken it directly.

Falsifier A

A four-regime architecture performs equally well after complexity penalty and exhibits no Constitution–Judgment Collapse.

Falsifier B

Closure and Selection cannot be dynamically or interventionally distinguished.

Falsifier C

Retention provides no advantage over ordinary accepted-memory storage.

Falsifier D

Residual preservation does not improve recovery after declaration shift.

Falsifier E

Blind switching models consistently infer another K.

Falsifier F

The five proposed modes fail to transfer across domains.

A serious theory should welcome these outcomes.

They tell us which distinctions were artifacts of description rather than properties of the system.

68. Where the Argument Now Stands

The article has now reached a point where the five-regime structure is no longer merely a suggestive classification.

It has:

- a target class of systems;
- five candidate functional necessities;
- distinct objectives;
- characteristic failure modes;
- a hybrid-system representation;
- dual trace–residual ledgers;
- residual-driven transition pressure;
- hysteresis;
- nested revision depth;
- explicit four- and six-regime competitors;
- empirical falsifiers.

That is enough to treat the five-regime structure as a genuine research hypothesis.

What remains is to connect it to concrete experimental programs and then, only afterward, discuss the striking historical analogy with the traditional Five Phases.

---

69. Next Step: From Kernel to Experiments

The next sections should answer three questions.

First:

> How can this architecture be implemented in an artificial agent without simply hard-coding the desired result?

Second:

> What experiments distinguish five regimes from four or six?

Third:

> What novel predictions follow specifically from the residual-ledger and world-revision mechanism?

Only after those sections should the article introduce the historical Wuxing comparison.

That ordering will make the eventual cultural correspondence much more convincing because the reader will already have seen the engineering theory stand on its own.

70. Experimental Realization Without Hard-Coding Five

The most dangerous experiment would be to build five labeled modules called Generation, Activation, Closure, Selection, and Retention and then report that the resulting system exhibits

five phases.

That would demonstrate only the architecture we inserted.

A stronger experiment should hard-code only the **functional pressures** required by a persistent self-revising bounded system.

The system should receive:

- a nonstationary environment;
- a finite observation budget;
- an operational world-model;
- a memory budget;
- a cost for revising that world-model;
- an unresolved-residual channel;
- the ability to propose alternative models.

The controller should then learn its own operating regimes.

Let the latent control state be:

$z_t \in \{1,\ldots,K\}$. (70.1)

Neither K nor the semantics of the modes should initially be specified.

The first experiment therefore asks:

$K^* = \text{argmax}_K\, Q(M_K)$. (70.2)

Only after K^* has been estimated should the learned modes be interpreted.

This protects the central hypothesis from one of the most obvious sources of confirmation bias.

---

71. A Minimal Artificial Environment

A useful test environment should periodically change in ways that cannot be handled by simple parameter adjustment.

Suppose observations are generated from:

$y_t \sim P(y \mid x_t, D^*_t)$. (71.1)

The true latent regime D^*_t occasionally changes:

$D_n \rightarrow D_{n+1}$. (71.2)

Some changes should be shallow.

They can be absorbed through:

$x \rightarrow x'$. (71.3)

Others should alter the relevant variables or causal structure itself and therefore require:

$D \rightarrow D'$. (71.4)

The agent does not receive a label indicating which kind of change occurred.

It must infer whether to:

- adapt;
- explore;
- test;
- recommit;
- prune;
- consolidate;
- revise its world.

This environment directly tests the distinction between **state failure** and **declaration failure**.

---

72. The Crucial Experimental Control

A standard adaptive agent should be compared against a residual-preserving agent.

Baseline architecture

Prediction error is immediately optimized away:

$e_t \rightarrow$ parameter update. (72.1)

Residual-ledger architecture

Mismatch is divided into:

$e_t \rightarrow$ locally absorbable component + unresolved residual. (72.2)

The unresolved component is preserved:

$\bar{L}_{t+1} = \bar{L}_t \oplus r_t$. (72.3)

The central test is whether the second architecture becomes better at distinguishing:

“the state is wrong”

from

“the world-model is wrong.”

If residual preservation provides no advantage, one of the strongest proposed SMFT mechanisms would be weakened.

---

73. Prediction 1 — Directional Residual Should Precede Deep Revision

The framework predicts that declaration change should not correlate only with raw error magnitude.

A better predictor should combine magnitude with directional persistence.

Thus:

$P(D_{n+1} \neq D_n \mid \rho, \delta R) > P(D_{n+1} \neq D_n \mid \rho)$. (73.1)

In words:

> residual direction should carry information about coming world revision beyond residual magnitude alone.

This is a strong prediction because it can fail.

If magnitude alone performs equally well, the directional-residual construction adds unnecessary machinery.

---

74. Prediction 2 — Revision Should Exhibit Hysteresis

If declarations are genuinely latched rather than continuously re-estimated, transition thresholds should depend on direction.

Let:

Θ_{leave} = residual pressure required to abandon D_1,

and:

Θ_{return} = residual pressure below which D_1 becomes attractive again.

The theory predicts:

$\Theta_{leave} \neq \Theta_{return}$. (74.1)

This should produce measurable hysteresis loops.

The mature SMFT work already treats Gate/Latch behavior and world revision as distinct from ordinary continuous adaptation; the present architecture turns that into an explicit experimental prediction rather than leaving it at the conceptual level.
[探討 SMFT 串聯其它名門正派成為五行流轉的可行性]

---

75. Prediction 3 — Deep Revision Should Be Rarer Than State Repair

Because revision depth carries increasing cost:

$C_{state} < C_{regime} < C_{declaration} < C_{purpose}$. (75.1)

the frequency of successful updates should approximately satisfy:

$N_{state} > N_{regime} > N_{declaration} > N_{purpose}$. (75.2)

A system that continuously rewrites its ontology or identity is not demonstrating superior adaptability.

It is demonstrating failure to stabilize.

This prediction separates **adaptive depth** from **revision frequency**.

---

76. Prediction 4 — Residual Preservation Should Improve Recovery After Ontology Shift

Suppose a previously rejected hypothesis becomes useful after the environment changes.

An ordinary optimizer may have erased its trace.

A dual-ledger system should retain enough of the previous anomaly structure to recover it more quickly.

Define recovery latency:

Trec = time required to regain adequate performance after a declaration shift. (76.1)

Then the residual-ledger hypothesis predicts:

E[Trec | dual ledger] < E[Trec | admitted-only memory]. (76.2)

This may be one of the cleanest empirical tests in the entire framework.

---

77. Prediction 5 — Functional Ablations Should Fail Differently

Five-regime theory predicts not merely reduced performance when one function is removed, but **qualitatively different failure profiles**.

Remove Generation:

→ adaptation remains local and increasingly rigid.

Remove Activation:

→ candidate worlds proliferate without adequate causal testing.

Remove Closure:

→ useful local behaviors exist without stable global organization.

Remove Selection:

→ complexity and contradiction accumulate.

Remove Retention:

→ adaptation may occur rapidly but continuity and recovery deteriorate.

If all five ablations merely reduce one generic score by similar amounts, the decomposition has little explanatory value.

---

78. Prediction 6 — Mode Switching Should Matter More Than Mode Capacity

A sufficiently large neural system may contain all five functional capacities simultaneously.

The crucial question is therefore not:

> Does the system possess the five abilities?

It is:

> Can it allocate control dominance appropriately among them?

Define transition error:

$$E_switch = \Sigma_t \, 1[q_t \neq q^*_t]. \quad (78.1)$$

A system with strong individual sub-capabilities but poor switching may underperform a smaller system with better regime control.

This leads to an important architectural prediction:

> **Meta-control over cognitive regimes may become more important than simply increasing competence within each regime.**

---

79. A Possible AI Architecture

One implementation could contain a shared latent model plus specialized control policies.

Let:

$$h_t = \text{shared latent world state.} \quad (79.1)$$

Each regime has a policy:

$$\pi G(a \mid h, L^+, L^-),$$

$$\pi A(a \mid h, L^+, L^-),$$

$$\pi C(a \mid h, L^+, L^-),$$

$$\pi S(a \mid h, L^+, L^-),$$

$$\pi R(a \mid h, L^+, L^-). \quad (79.2)$$

A meta-controller chooses:

$$q_t \sim \pi meta(q \mid h, L^+, L^-, P). \quad (79.3)$$

The model need not literally contain five independent neural networks.

They may be:

- low-rank adapters;
- routing heads;
- recurrent control states;
- mixture-of-expert policies;
- dynamically selected objective functions;
- latent metastable modes.

The theory concerns functional organization, not implementation identity.

---

80. Another Implementation: Continuous Controller Coordinates

The five modes need not be discrete.

Let:

$\alpha(t) = (\alpha G, \alpha A, \alpha C, \alpha S, \alpha R)$. (80.1)

with:

$\alpha q \geq 0, \quad \Sigma q \, \alpha q = 1$. (80.2)

Then:

$\dot{X} = \Sigma q \, \alpha q \, Fq(X)$. (80.3)

A strongly phase-separated system will spend most of its time near vertices of the simplex.

For example:

$\alpha \approx (1,0,0,0,0)$

indicates Generation dominance.

A less phase-separated system may occupy mixtures.

The empirical question becomes:

> Does the learned control trajectory develop five metastable corners?

This formulation avoids forcing categorical transitions where the underlying system is continuous.

---

81. Metastability Is More Important Than Hard Discreteness

The Five-Regime Modal Hypothesis does not require perfectly discrete states.

A regime may instead correspond to a metastable region:

B_G, B_A, B_C, B_S, B_R. (81.1)

The controller trajectory $\alpha(t)$ spends long periods near one basin and occasionally transitions between basins.

The relevant observable becomes a transition matrix:

$Pij = P(q_{t+1}=j \mid q_t=i)$. (81.2)

The canonical circulation hypothesis predicts enhanced probability along:

G→A,

A→C,

C→S,

S→R,

R→G,

relative to appropriate null models.

Again, not probability one.

---

82. The Important Result Is Not Necessarily K=5

Suppose experiments repeatedly infer:

K=4.

That would not destroy the entire boundary-formation framework.

It would suggest that two proposed functions belong to one metastable regime.

Likewise:

K=6

would suggest that one current regime contains a hidden independently stable function.

The important theoretical advance would still remain:

> regime number has become a measurable property rather than a symbolic assumption.

This distinction should be made explicit throughout the article.

---

83. From Five Regimes to Signed Interaction Topology

There is now a deeper question.

So far we have studied the **successive handoff** between regimes.

But robust systems require more than succession.

Each regime must also prevent another regime from becoming pathological.

This introduces a second type of relation.

Let:

$E^{+}ij > 0$

denote an enabling or productive coupling.

Let:

$E^{-}ij < 0$

denote a restraining or counterbalancing coupling.

The architecture may therefore contain both:

productive flow

and

cross-regime regulation.

This turns the five-regime model from a simple cycle into a signed control network.

---

84. The Productive Ring

The proposed productive relationships are:

$G \to A$ generated possibility becomes enactable. (84.1)

$A \to C$ successful enactment creates pressure for integration. (84.2)

$C \to S$ commitment creates an accountable structure for evaluation. (84.3)

$S \to R$ evaluation determines what should be consolidated or preserved. (84.4)

$R \to G$ retained history and residual seed new possibility. (84.5)

This produces:

$G \to A \to C \to S \to R \to G.$ (84.6)

Call this the **productive ring**.

---

85. A Second Ring Appears From Failure Prevention

Now ask a different question:

> Which regime most naturally restrains the pathological excess of another?

The answer is unexpectedly structured.

---

86. Selection Restrains Generation

Generation expands candidate variety.

Unrestrained Generation produces:

- combinatorial explosion;
- incoherence;
- perpetual novelty.

Selection counters this by pruning low-value candidates.

Thus:

$S \dashv G.$ (86.1)

where ⊣ denotes functional restraint.

This relation is straightforward.

---

87. Retention Restrains Activation

Activation raises causal activity, resource expenditure, and possible amplification.

Retention introduces:

- cooling;
- latency;
- consolidation;
- recovery;
- withdrawal from active processing.

Thus:

$R \dashv A.$ (87.1)

A system incapable of this transition burns continuously.

Retention therefore counterbalances runaway Activation.

---

88. Generation Restrains Closure

Closure stabilizes a declared world.

Excessive Closure becomes:

- dogmatism;
- lock-in;
- inability to reinterpret anomalies.

Generation introduces alternative distinctions and candidate worlds.

Thus:

$G \dashv C.$ (88.1)

Generation is not simply what occurs before Closure.

It is also one of the forces capable of **breaking excessive Closure**.

This makes the interaction network nontrivial.

---

89. Closure Restrains Retention

Retention preserves:

- history;
- dormant alternatives;

- unresolved residual;
- latent possibilities.

But unlimited retention is impossible for a bounded system.

A declaration supplies:

- identity constraints;
- admissibility criteria;
- memory schemas;
- compression boundaries;
- resource priorities.

Closure therefore limits what can remain indefinitely available.

Thus:

$C \dashv R.$ (89.1)

This relation should not mean “delete the residual.”

It means that retention must itself occur inside some bounded structure.

Without Closure, the system becomes an indiscriminate archive.

---

90. Activation Restrains Selection

This edge is subtler.

Excessive Selection produces:

- analysis paralysis;
- premature pruning;
- excessive conservatism;
- rejection of candidates before sufficient evidence exists.

Activation counterbalances this by forcing candidates into consequential interaction.

A candidate that survives only abstract criticism but is never enacted cannot produce decisive evidence.

Thus:

$A \dashv S.$ (90.1)

Activation functions here as a restraint on over-selection:

> stop pruning indefinitely; test something.

This completes a second cycle.

---

91. The Cross-Regulatory Ring

Collecting the five restraining relations:

$G \dashv C$, (91.1)

$C \dashv R$, (91.2)

$R \dashv A$, (91.3)

$A \dashv S$, (91.4)

$S \dashv G$. (91.5)

Therefore:

$G \dashv C \dashv R \dashv A \dashv S \dashv G$. (91.6)

This structure was **not needed to derive the five primary functions**.

It appears only after asking how pathological overexpression of each function might be counterbalanced.

That makes the result unusually interesting.

---

92. A Signed Interaction Matrix

Order the modes as:

(G,A,C,S,R). (92.1)

A schematic signed interaction matrix is:

	G	A	C	S	R
G	o	+	−	o	o
A	o	o	+	−	o
C	o	o	o	+	−
S	−	o	o	o	+
R	+	−	o	o	o

(92.2)

Here:

- denotes productive handoff,

– denotes cross-regulatory restraint,

o denotes a relation not yet specified at this coarse level.

This matrix should not yet be treated as empirically established.

It is a compact conjectural control topology.

---

93. A More General Dynamical Form

Let aq denote regime activation.

Then:

$\dot{a}_i = a_i[F_i(S) - \Sigma_j K_{ij} a_j]$. (93.1)

The matrix K can contain both cooperative and inhibitory interactions.

One may alternatively write:

$\dot{a} = \mathrm{diag}(a)[b + W a - \varphi(a)]$. (93.2)

where W is a signed coupling matrix.

The theory predicts that a useful learned W may exhibit enhanced weights corresponding approximately to:

productive neighbour couplings

plus

cross-regulatory skip couplings.

This provides another experimental route.

---

94. Why the Double-Cycle Structure Matters

A one-cycle model explains only:

> how a healthy process progresses.

A two-cycle model additionally explains:

> how each healthy process prevents another healthy process from becoming pathological.

That is a much richer architecture.

For example:

Generation is beneficial.

Too much Generation is destabilizing.

Selection restrains it.

But too much Selection is itself pathological.

Activation restrains excessive Selection.

And so on.

The system therefore contains no universally dominant virtue.

Each function is necessary.

Each function becomes dangerous when unconstrained.

This is precisely the kind of architecture one expects in a persistent adaptive system.

---

95. A Balance Condition

Let aq denote normalized regime intensity.

A crude homeostatic condition is:

$\Sigma q\ aq = 1.$ (95.1)

But equal activation is not required.

More interesting is the condition that no regime's effective gain exceed the regulatory capacity of the others for too long.

Define:

Γq = productive gain of q − regulatory load on q. (95.2)

Healthy circulation may require:

$\langle \Gamma q \rangle_T \approx 0$ over sufficiently long windows. (95.3)

Short excursions are allowed.

Persistent positive Γq means runaway dominance.

Persistent negative Γq means suppression or functional extinction.

This begins to give “balance” an engineering meaning.

---

96. Why a Five-Node Network Is Geometrically Compact

There is an interesting graph-theoretic feature.

Place the five regimes on the productive cycle:

G — A — C — S — R — G.

The productive relations connect neighbouring nodes.

The cross-regulatory relations connect the non-neighbouring nodes.

For five nodes, these two edge sets form:

- an outer five-cycle;
- an inner five-cycle.

Together, as undirected relations, they exhaust all ten pairs of the complete graph K_5.

This is an exact graph fact.

It means that once five modes exist, the distinction:

“productive neighbour”

versus

“cross-regulating non-neighbour”

provides a particularly compact decomposition of all pairwise relationships.

This does **not** prove that there must be five regimes.

But it gives the five-regime architecture an additional mathematical economy.

---

97. The Pentagonal Control Graph

Let:

C_5 = productive cycle. (97.1)

Let:

C_5^* = skip-one regulatory cycle. (97.2)

Then:

$E(K_5) = E(C_5) \cup E(C_5^*)$, (97.3)

with:

$E(C_5) \cap E(C_5^*) = \emptyset$. (97.4)

Thus:

$K_5 = C_5 \oplus C_5^*$. (97.5)

Here $\oplus$ means edge-disjoint union.

This provides a remarkably compact representation:

> every pair of regimes can be classified either as a neighbouring productive relation or as a cross-regulatory relation.

Again, the graph structure follows **after** five has been hypothesized.

It must not be used as a reverse proof of five.

---

98. A New Conjecture: Dual-Cycle Boundary Regulation

We can now state a stronger hypothesis.

> **Dual-Cycle Boundary Regulation Conjecture.**
> If five functional regimes form a useful coarse-graining of persistent self-revising control, their stable interaction topology may contain two complementary cycles: a productive handoff cycle linking successive constructive functions, and a cross-regulatory cycle in which each regime counterbalances the pathological excess of a nonadjacent regime.

In compact form:

$G \to A \to C \to S \to R \to G$, (98.1)

and:

G ⊣ C ⊣ R ⊣ A ⊣ S ⊣ G. (98.2)

This is currently a conjecture.

But it is a much stronger and more interesting conjecture than merely saying that there are five stages.

---

99. This Produces New Failure Predictions

The dual-cycle model predicts two fundamentally different kinds of pathology.

Productive-edge failure

An expected handoff does not occur.

For example:

G ↛ A

produces sterile ideation.

A ↛ C

produces endless experimentation without integration.

Regulatory-edge failure

A restraining relation is lost.

For example:

S ↛ G restraint

produces uncontrolled novelty.

R ↛ A restraint

produces runaway activation.

G ↛ C restraint

produces rigid closure.

This allows diagnosis of failures that would look identical in a simple sequential model.

---

100. Edge Ablation Experiments

Instead of ablating entire regimes, one can now ablate individual couplings.

Examples:

Remove S ⊣ G.

Prediction:

candidate diversity rises but useful conversion declines.

Remove R ⊣ A.

Prediction:

activation episodes become longer, recovery slows, resource fatigue rises.

Remove G ⊣ C.

Prediction:

declarations become increasingly persistent under ontology shift.

Remove C ⊣ R.

Prediction:

memory and residual accumulation become less bounded and less compressible.

Remove A ⊣ S.

Prediction:

selection becomes increasingly conservative and exploration is pruned before empirical enactment.

These predictions make the network experimentally richer.

---

101. The Historical Comparison Can Now Be Introduced

Only at this stage does an old fivefold process model become relevant.

Traditional Chinese thought developed the concept of **Wuxing (**五行**)**.

A literal translation of *xing* as “element” is misleading in this context.

For the present comparison, **Five Phases**, **Five Processes**, or **Five Movements** is closer to the functional interpretation required here.

The conventional sequence is:

Wood → Fire → Earth → Metal → Water → Wood.

The traditional system also contains a second regulatory sequence, usually described as a controlling or overcoming cycle.

We have deliberately not used either sequence to derive the engineering model above.

That methodological separation matters. The mature SMFT audit explicitly warns against letting comparative traditional structures reverse-generate the Core theory; such correspondences belong at the comparative layer rather than serving as evidence for the underlying mathematics. 𝕆 → G₂_SO(4) → ℍ → ℂ² 成界過程初探 1…

---

102. The First Correspondence

The independently proposed five regimes map naturally as:

Wood — Generation / Differentiation

Fire — Activation / Amplification

Earth — Closure / Integration

Metal — Selection / Refinement

Water — Retention / Latency

Thus:

Wood → Fire → Earth → Metal → Water → Wood

corresponds to:

G → A → C → S → R → G. (102.1)

The match is exact at the level of cyclic ordering.

This should be described carefully.

It is **not evidence that ancient terminology predicted modern AI architecture**.

It is evidence that two independently formulated functional decompositions possess an interesting structural resemblance.

---

103. Why the Mapping Is Not Merely Lexical

The correspondence is stronger if each label captures a functional tendency rather than a poetic adjective.

Wood / Generation

branching, emergence, differentiation, new structure.

Fire / Activation

expression, propagation, amplification, active consequence.

Earth / Closure

integration, stabilization, common ground, bounded organization.

Metal / Selection

cutting, discrimination, refinement, reduction of unsupported structure.

Water / Retention

storage, latency, cooling, persistence, return potential.

The important point is not whether every traditional description fits perfectly.

The question is whether the **functional topology** remains meaningful under modern engineering definitions.

---

104. A More Surprising Correspondence: The Regulatory Cycle

Traditional Wuxing also contains the **Ke cycle (克)**, often translated as the controlling, restraining, or overcoming cycle.

Its conventional order is:

Wood ⊣ Earth, (104.1)

Earth ⊣ Water, (104.2)

Water ⊣ Fire, (104.3)

Fire ⊣ Metal, (104.4)

Metal ⊣ Wood. (104.5)

Now compare the cross-regulatory relations independently proposed above:

Generation ⊣ Closure, (104.6)

Closure ⊣ Retention, (104.7)

Retention ⊣ Activation, (104.8)

Activation ⊣ Selection, (104.9)

Selection ⊣ Generation. (104.10)

Under the tentative correspondence:

G ↔ Wood

A ↔ Fire

C ↔ Earth

S ↔ Metal

R ↔ Water

the two regulatory cycles have the same topology.

This is substantially more interesting than merely matching five labels.

---

105. Why This Result Deserves Attention

If we had begun with Wuxing and invented five engineering functions to reproduce its two cycles, the result would have little evidential value.

But the reasoning proceeded in the opposite direction.

First:

we decomposed the requirements of a persistent self-revising system.

Then:

we obtained a candidate productive cycle.

Then:

we asked what restrains the pathological excess of each regime.

Only then did a second cross-regulatory cycle appear.

The fact that both cycles correspond to the two classical Wuxing relations is therefore an **unexpected structural coincidence within the present derivation**.

It is not yet scientific confirmation.

But it is sufficiently nontrivial to justify further investigation.

---

106. The Correct Epistemic Status

The correspondence should be labeled:

[H] engineering five-regime hypothesis

[H] dual-cycle regulation hypothesis

[A] Wuxing structural correspondence

not:

[K] established scientific fact.

The old framework should not be allowed to rescue a failed engineering model.

If experiments show:

$K^*=4$,

or a different regulatory topology,

the modern model should change even if the traditional correspondence becomes less elegant.

That is the necessary firewall.

---

107. A Potentially Important New Interpretation of “Generation” and “Control”

The two Wuxing cycles are traditionally distinguished as something like:

Sheng (生) — generation / production

Ke (克) — control / restraint.

The present engineering model gives these terms an unusually precise interpretation.

Generative relation

$i \to j$

means:

successful output of regime i supplies useful conditions for regime j.

Regulatory relation

$i \dashv j$

means:

regime i suppresses the pathological overexpression of regime j.

This distinction can be formalized.

Let a_i denote activity of regime i.

Then:

$\partial \dot{a}_j / \partial a_i > 0$ for productive coupling. (107.1)

and:

$\partial \dot{a}_j / \partial a_i < 0$ for regulatory coupling. (107.2)

The ancient distinction is therefore translated into the language of **signed dynamical coupling**.

That is a legitimate engineering object.

---

108. The Double Ring as a Control Architecture

Define the productive adjacency matrix P and regulatory matrix K.

For node ordering:

(G,A,C,S,R),

let:

$P_{GA} = P_{AC} = P_{CS} = P_{SR} = P_{RG} = +1.$ (108.1)

and:

$K_{GC} = K_{CR} = K_{RA} = K_{AS} = K_{SG} = -1.$ (108.2)

Then a simplified regime interaction system is:

$\dot{a}_i = a_i[b_i + \Sigma_j(P_{ij} + K_{ij})a_j - \varphi_i(a_i)]$. (108.3)

One may then ask:

- does the double-ring network possess stable cycling solutions?
- under what gains does one mode dominate?
- where do oscillations arise?
- when does the system latch?
- when does regulation fail?
- how does residual perturb the cycle?

These are ordinary dynamical-systems questions.

No metaphysical commitment is required.

---

109. Residual as the Perturbation That Keeps the Cycle Alive

A perfectly closed cycle might simply repeat.

Persistent intelligence requires adaptation.

Residual can supply the perturbation.

Let:

R_n = accumulated unresolved structure after cycle n. (109.1)

Then allow regime gains to depend on residual:

$b_i = b_i(P,L^+,R_n)$. (109.2)

In particular:

bG increases when structured residual accumulates. (109.3)

This means that the next cycle is not identical to the previous one.

Residual deforms the control landscape.

The resulting process is:

cycle → trace/residual → landscape deformation → revised cycle. (109.4)

This is closer to recursive world formation than to a static periodic oscillator.

---

110. From Cycle to Spiral

The geometry should therefore be thought of as a spiral rather than a closed circle.

A pure cycle satisfies:

$S_{n+1} = S_n$. (110.1)

A self-revising cycle instead satisfies:

$S_{n+1} = \Phi(S_n, L^+_n, L^-_n)$. (110.2)

and generally:

$S_{n+1} \neq S_n$. (110.3)

Yet the functional ordering may remain recognizable.

Thus:

$G \to A \to C \to S \to R \to G' \to A' \to C' \to \ldots$ (110.4)

The prime indicates that the next Generation regime begins from a historically altered world.

This is more accurately a **recursive circulation**.

---

111. A Five-Regime Fractal Possibility

Because each regime can itself contain a smaller five-regime cycle, the architecture may be recursively nested.

At level ℓ:

$Q^{(\ell)} = \{G^{(\ell)}, A^{(\ell)}, C^{(\ell)}, S^{(\ell)}, R^{(\ell)}\}$. (111.1)

A regime at level ℓ may contain a complete subcycle at level $\ell+1$.

For example:

$S^{(macro)}$

may require:

$G^{(micro)} \to A^{(micro)} \to C^{(micro)} \to S^{(micro)} \to R^{(micro)}$. (111.2)

The result resembles a self-similar control grammar.

This possibility is especially relevant to SMFT because the broader theory already treats recursive disclosure and revision as fundamental organizational motifs.

It remains a hypothesis.

---

112. The Double-Cycle Model Suggests a New Kind of Balance

“Balance” need not mean equal quantities.

Instead define a dynamically balanced state as one in which:

1. every necessary regime remains reachable;
2. no regime permanently monopolizes control;

3. productive transitions remain available;
4. regulatory counter-couplings remain strong enough to prevent runaway behavior;
5. residual can eventually alter the regime landscape.

Thus balance means:

preserved circulation capacity.

This is a much sharper concept than static equilibrium.

---

113. Balance Versus Equilibrium

A balanced self-revising system may never be at equilibrium.

Indeed:

$\dot{X} \neq 0$

can remain true indefinitely.

What remains stable is the capacity for appropriate circulation.

Let:

$\mathcal{C}irc(S)$ = accessibility of required regime transitions. (113.1)

Healthy operation requires:

$\mathcal{C}irc(S) > \mathcal{C}min$. (113.2)

A pathological system may have:

$\dot{X} \approx 0$

while also:

$\mathcal{C}irc(S) \approx 0$.

That is not healthy equilibrium.

It is lock-in.

---

114. The Difference Between Stability and Vitality

This suggests a useful distinction.

Stability

the current state resists perturbation.

Vitality

the system retains the capacity to enter the regime required by future conditions.

A highly closed declaration may be stable yet non-vital.

A highly generative system may be flexible yet unstable.

Persistent intelligence requires both:

stability when appropriate

and

transition capacity when necessary.

The five-regime architecture offers a way to formulate that tradeoff.

---

115. A Regime-Reachability Measure

Let the regime transition graph under current system conditions be G_S.

Define reachability:

Reach(i,j | S)=1

if regime j is accessible from regime i under admissible interventions.

Then define:

$V_regime = \Sigma_{i \neq j} Reach(i,j \mid S).$ (115.1)

Low V_regime indicates that the system has lost functional flexibility.

This may provide an operational definition of one aspect of adaptive vitality.

---

116. Pathologies as Broken Circulation

Many familiar system failures can now be interpreted as circulation defects.

Exploration lock

R → G fails.

The system preserves history but cannot reopen possibility.

Activation lock

A cannot transition to C or R.

The system remains permanently engaged.

Closure lock

C resists G and S excessively.

The world becomes dogmatic.

Selection lock

S suppresses both G and A.

The system becomes chronically critical but unproductive.

Retention failure

S → R fails.

The system continually re-processes the same material because nothing consolidates.

This turns qualitative pathology into transition topology.

---

117. A New Perspective on Intelligence

Under this framework, intelligence is not simply:

prediction,

reasoning,

planning,

or optimization.

A deeper operational criterion may be:

> **the ability to allocate the correct kind of transformation to the correct depth of failure while preserving enough historical continuity to revise without disintegrating.**

This includes:

knowing when to generate,

when to act,

when to commit,

when to criticize,

when to stop,

and when accumulated failure means that the world itself must change.

That is a richer definition of adaptive competence.

---

118. Why This Matters for AGI

As systems become more autonomous and persistent, fixed-context competence becomes less sufficient.

A long-lived agent will encounter:

- changing environments;
- conflicting tasks;

- outdated ontologies;
- broken abstractions;
- changing tools;
- adversarial perturbations;
- memory accumulation;
- self-generated commitments.

The crucial problem becomes:

> How can such an agent remain revisable without becoming unstable?

The five-regime architecture offers one candidate answer.

Not:

continuous self-modification.

But:

governed self-revision through functionally distinct regimes, residual accounting, latching, and revision-depth control.

---

119. Implication for AI Safety

The architecture suggests that safe adaptability may require both:

revision capacity

and

revision resistance.

Too little revision produces rigidity.

Too much revision destroys continuity.

Safety therefore cannot be reduced to maximizing either stability or flexibility.

One needs controlled thresholds between:

local correction,

regime shift,

world revision,

and purpose revision.

This gives safety a dynamical form.

---

120. Rollback and Counterfactual Recovery

The residual ledger also supports rollback.

If previous structures are not erased completely, a system can retain:

- rejected hypotheses;
- previous declarations;
- transition reasons;
- residuals that motivated revision.

Let:

$H_D = \{D_0, D_1, \ldots, D_n\}$. (120.1)

Then a failed revision can permit:

$D_{n+1} \to D_k$

for some $k<n$,

provided historical compatibility remains sufficient.

This gives self-revision a recoverable structure rather than a one-way overwrite.

---

121. Memory Should Store Structural History, Not Only Content

Traditional memory systems often preserve:

documents,

vectors,

episodes,

facts.

A self-revising architecture may also need to store:

- which declaration admitted the trace;
- which regime produced it;
- what was excluded;
- why a revision occurred;
- what alternative declarations were rejected.

A richer memory object is therefore:

$M_t = (T_t, r_t, D_t, q_t, \Gamma_t)$. (121.1)

where Γ_t contains transition or provenance metadata.

Memory then becomes part of the system's structural accountability.

---

122. The Observer Is Also Inside the Cycle

The observer should not be modeled as an external auditor.

Observation depends on:

D,

P,

historical trace,

and current regime.

Thus:

$O_t = O(D_t, P_t, L^+_t, L^-_t, q_t)$. (122.1)

The observer that performs Selection may not be identical, functionally, to the observer active during Generation.

This suggests that the five-regime architecture may also imply **regime-dependent observation policies**.

That is an important extension.

---

123. Regime-Dependent Observation

During Generation, the observer may prioritize:

novelty and anomaly.

During Activation:

causal consequence.

During Closure:

coherence and compatibility.

During Selection:

discrimination and counterexample.

During Retention:

compressibility, stability, and future recoverability.

Therefore:

$\Pi q \neq \Pi q'$ in general. (123.1)

The system does not merely change what it does.

It may change **what it looks for**.

This greatly increases the potential importance of regime control.

---

124. A Stronger Self-Referential Loop

Because observation changes by regime:

q → observation policy → trace/residual → ledger → revision pressure → q′. (124.1)

Thus the regime controller partly determines the evidence that later determines the regime controller.

This is a self-referential loop.

It needs governance.

Otherwise a mode can create evidence favouring its own continuation.

For example:

Selection mode may preferentially observe defects,

thereby generating more evidence for continued Selection.

This creates mode lock-in.

---

125. Anti-Lock Mechanisms

One possible safeguard is mandatory cross-regime audit.

After prolonged occupancy:

T_dwell(q) > T_max(q), (125.1)

the system invokes an alternative observer.

For example:

Selection-dominant system → Generation audit.

Closure-dominant system → residual challenge.

Activation-dominant system → Retention audit.

This makes the regulatory cycle operational rather than metaphorical.

---

126. The Regulatory Cycle as Anti-Lock Governance

The cross-regulatory edges can therefore be interpreted as **anti-lock channels**.

S ⊣ G:

Selection prevents generative explosion.

G ⊣ C:

Generation prevents closure lock.

C ⊣ R:

Closure prevents unbounded latent accumulation.

R ⊣ A:

Retention prevents activation lock.

A ⊣ S:

Activation prevents selection lock.

The double-cycle topology may therefore implement a distributed anti-monopoly system for cognition.

No regime can safely govern forever.

---

127. A Governance Interpretation

Let each regime possess local authority A_q.

A healthy architecture prevents:

$A_q \to 1$ permanently. (127.1)

The regulatory network ensures that another regime retains a legitimate intervention path.

This resembles separation of powers in governance:

not because the regimes correspond to political institutions,

but because robustness arises when no single objective controls both:

evidence generation,

admissibility,

evaluation,

and memory.

This analogy may later be useful, though it is not required for the formal theory.

---

128. What Five Adds Beyond a Generic Explore–Exploit Model

A common objection is that adaptive systems already distinguish exploration and exploitation.

The present framework contains that distinction but goes much further.

Exploration corresponds only partially to G.

Exploitation corresponds only partially to A or C.

The five-regime model additionally distinguishes:

- enactment from commitment;
- commitment from criticism;
- criticism from retention;
- retained anomaly from erased error;
- state adaptation from world revision.

Therefore:

Five-Regime Cycle ≠ Explore–Exploit with extra terminology. (128.1)

It addresses a deeper level of organization.

---

129. What Five Adds Beyond OODA-Like Loops

Another objection is that many control frameworks already cycle through:

observe,

orient,

decide,

act.

Such loops are valuable but generally assume a relatively stable frame within which observation and decision occur.

The present architecture asks an additional question:

> What happens when the frame defining observation, orientation, decision, and action becomes inadequate?

The answer requires:

declaration,

residual preservation,

and world revision.

That is the distinctive problem.

---

130. What Five Adds Beyond Error-Correction

Error-correction assumes a code or model against which deviations are measured.

SMFT-style residual dynamics asks:

> What if repeated “error” is evidence that the code itself should change?

Thus:

Error correction = restore state under D. (130.1)

World revision = revise D under persistent structured residual. (130.2)

This distinction should remain central.

---

131. The Strongest Current Form of the Theory

We can now state the architecture in one sentence:

> A persistent self-revising bounded system maintains an effective world through five potentially metastable control regimes—Generation, Activation, Closure, Selection, and Retention—while a dual trace–residual ledger and hysteretic switching mechanism determine when ordinary adaptation is insufficient and deeper world revision is required.

The stronger dual-cycle extension adds:

> Productive handoff follows G→A→C→S→R→G, while cross-regulatory restraints follow G⊣C⊣R⊣A⊣S⊣G.

That is now a compact theoretical object.

---

132. Historical Correspondence: What Can Legitimately Be Claimed

The independent derivation produces a striking correspondence with Wuxing:

Wood ↔ Generation

Fire ↔ Activation

Earth ↔ Closure

Metal ↔ Selection

Water ↔ Retention.

Both the productive and regulatory topologies correspond to the traditional Sheng (生, generative) and Ke (克, controlling) cycles.

The strongest legitimate statement is therefore:

> An independently motivated five-regime control architecture for persistent self-revising systems exhibits a structural isomorphism, at the level of its proposed two-cycle interaction graph, with the traditional Wuxing generative and controlling topology.

Even this statement should be qualified.

“Isomorphism” here refers to the abstract directed five-node topology under the proposed mapping.

It does not imply equivalence of ontology, historical meaning, or scientific status.

---

133. What Must Not Be Claimed

The present argument does **not** establish that:

- Wuxing is a scientific theory of AI;
- ancient authors knew modern control theory;
- every adaptive system must have five phases;

- the five regimes have been empirically observed;
- the Wuxing correspondence proves SMFT;
- the five AI research traditions must each occupy exactly one phase.

Those would all exceed the evidence.

The existing SMFT audit explicitly requires traditional correspondences to remain comparative interpretations rather than reverse proofs of the Core. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

---

134. What the Correspondence *Does* Justify

It justifies a research question:

> Why does an old qualitative process grammar and a modern independently constructed self-revision architecture admit the same two complementary five-cycle topologies?

Possible answers include:

1. coincidence;
2. flexible reinterpretation;
3. generic properties of five-node cyclic systems;
4. recurring human observation of adaptive processes;
5. a deeper control-theoretic regularity.

The present theory does not choose among them.

That choice requires further work.

---

135. A Necessary Anti-Numerology Test

To prevent retrofitting, repeat the entire analysis with:

K=4,

K=6,

K=7.

For each K, ask whether an equally natural pair of:

productive cycle

and

regulatory cycle

emerges.

Then compare:

- functional independence;
- explanatory compression;
- stability;

- intervention signatures;
- graph economy.

Five should be preferred only if it wins for reasons that do not depend on knowing the traditional answer.

This continues the anti-numerological discipline already established in the earlier SMFT audit. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1...

---

136. A Particularly Interesting Graph Test

For K=5:

productive neighbour relations

plus

non-neighbour regulatory relations

exhaust all unordered node pairs.

For K>5, nearest-neighbour plus one skip distance does not generally exhaust all pairwise interactions.

This gives K=5 a compact graph-theoretic property.

It should not be confused with necessity.

But it motivates the question:

> Does a five-regime controller achieve unusually high interaction coverage with unusually low rule complexity?

That can be measured.

---

137. Interaction Compression Score

Define:

I(K) = number of relevant pairwise relationships explained by the rule set. (137.1)

Let:

R(K) = number of independent coupling rules required. (137.2)

Define compression:

C_int(K) = I(K) / R(K). (137.3)

A five-regime double-cycle architecture may exhibit high C_int because two simple cyclic rules organize every pairwise relation.

This is another quantitative hypothesis worth testing.

---

138. Beyond Five: The Theory Must Remain Revisable

The irony would be severe if a theory of self-revising worlds became dogmatically attached to five regimes.

The framework should therefore preserve its own residual.

If experiments repeatedly produce:

four,

six,

or task-dependent K,

the theory should revise.

The article should make this explicit.

Its central commitment is not to five.

It is to:

> **functional decomposition under residual-preserving revision.**

Five is the present best candidate.

---

139. A Better Name for the Hypothesis

At this stage, I would name the central proposal:

The Five-Regime Boundary Circulation Hypothesis

with the stronger network extension:

The Dual-Cycle Boundary Regulation Hypothesis

The first concerns the five functional regimes.

The second concerns the productive and regulatory topology between them.

The distinction is useful because the first could survive even if the second fails.

---

140. The Minimal Mathematical Kernel

The article can now compress the proposal into:

$$S = (D,P,x,L^{+},L^{-},q). \quad (140.1)$$

$$\dot{x} = f_q(x;D,P,L^{+},L^{-}). \quad (140.2)$$

$$(T,r) = \Pi_q,D,P,O(x). \quad (140.3)$$

$L^+ \leftarrow L^+ \oplus T.$ (140.4)

$L^- \leftarrow L^- \oplus r.$ (140.5)

$\Pi rev = \mathscr{R}(D,P,L^+,L^-).$ (140.6)

$q' = Q(q,S,\Pi rev).$ (140.7)

$D' = U(D,P,L^+,L^-)$ when revision guard is crossed. (140.8)

with:

$q \in \{G,A,C,S,R\}.$ (140.9)

This is probably sufficient mathematical machinery for the conceptual paper.

More detailed realizations can come later.

---

141. A Compact Signed Regime Law

If we want one equation representing the dual-cycle controller, use:

$\dot{a}_i = a_i[b_i(R,P,L) + \Sigma_j W_{ij}a_j - \varphi_i(a_i)].$ (141.1)

where:

$W = W^+ + W^-.$ (141.2)

W^+ encodes:

$G \to A \to C \to S \to R \to G.$ (141.3)

W^- encodes:

$G \dashv C \dashv R \dashv A \dashv S \dashv G.$ (141.4)

This equation is a construction, not a derivation.

But it gives the hypothesis a clear simulation target.

---

142. Stability Questions

The natural mathematical questions include:

1. Does the signed network admit stable fixed points?
2. Does it admit stable limit cycles?
3. Under what coupling strengths does winner-take-all domination appear?
4. When do hysteretic declaration transitions occur?
5. Can residual force controlled escape from a stable attractor?
6. Which interaction patterns maximize recovery after structural shocks?
7. Does K=5 provide superior robustness or compression to K=4 or K=6?

These are now well-posed research problems.

---

143. Simulation Should Come After the Structural Paper

The current article does not need to answer all these questions numerically.

Its contribution can be:

- define the problem;
- derive the functional decomposition;
- formulate competing regime counts;
- construct the residual-driven hybrid architecture;
- derive the dual-cycle hypothesis;
- state falsifiable predictions.

That is enough for a theoretical article.

Simulation should be the next paper or companion experiment.

---

144. Relationship to the Wider SMFT Program

Within SMFT, the five-regime architecture should sit at the **Core / control-grammar level**, not be tied initially to the octonionic extension or any particular declaration manifold.

The broad SMFT development already separates the Core—Observer, Declaration, Purpose, Gate, Trace, Filtration, Residual, Latching, Revision—from mathematical extensions and comparative interpretations.
[探討 SMFT 串聯其它名門正派成為五行流轉的可行性]

The present proposal belongs between the Core and its engineering realization.

It asks:

> How should those operators be orchestrated over time?

The five-regime cycle is one candidate answer.

---

145. Why This May Be a Qualitative Advance for SMFT

Earlier SMFT work already possessed:

Declaration,

Gate,

Trace,

Ledger,

Residual,

Revision.

What was less explicit was a **control grammar governing when different kinds of**

cognitive work should dominate.

The present architecture adds:

functional regime separation

plus:

cross-regime governance.

That potentially turns a list of important operators into a **circulating runtime**.

This is a substantial conceptual change.

---

146. From Operator Inventory to Runtime Governance

An operator inventory says:

the system has G, A, C, S, R-like capacities.

A runtime says:

when G should dominate,

when G should yield to A,

when C should resist G,

when R should suppress A,

when residual should force re-entry into G.

The distinction resembles:

having functions in a software library

versus

having an operating system that schedules them.

The five-regime hypothesis is therefore best interpreted as an attempted **runtime kernel for self-revising world formation**.

---

147. A Possible Engineering Name

For implementation work, a neutral name may be useful:

Boundary Circulation Controller — BCC

with:

BCC-5 = five-regime model.

Then competing architectures can be:

BCC-4,

BCC-5,

BCC-6.

This avoids embedding Wuxing terminology in the experimental apparatus.

Only afterward can the traditional comparison be discussed.

---

148. BCC-5 State

A minimal implementation state is:

$S_BCC = (D,x,P,L^+,L^-,a).$ (148.1)

where:

$a \in \Delta^4$

is a five-component regime activation vector on the simplex.

The controller updates:

$a_{t+1} = F_BCC(a_t,S_t).$ (148.2)

The dominant regime is:

$q_t = \text{argmax_i } a_i.$ (148.3)

This provides a concrete experimental implementation.

---

149. BCC-5 Performance Criteria

A successful controller should minimize:

$J_total = J_task + \lambda_1 J_revision + \lambda_2 J_forgetting + \lambda_3 J_lock + \lambda_4 J_instability.$ (149.1)

where:

J_task = ordinary task loss

J_revision = unnecessary declaration-change cost

J_forgetting = loss of recoverable useful structure

J_lock = inability to revise under true ontology shift

J_instability = excessive regime/frame switching

This objective deliberately penalizes both:

too much stability

and

too much revision.

That is the central design tension.

---

150. The Most Important Benchmark

The decisive benchmark should repeatedly alternate between:

parameter shift

and

ontology shift.

For parameter shift:

D should remain stable.

For ontology shift:

D should eventually change.

The controller is rewarded for correctly distinguishing them.

Define:

TP_D = correct declaration revisions.

FP_D = unnecessary declaration revisions.

FN_D = missed declaration revisions.

Then:

Precision_D = TP_D / (TP_D + FP_D). (150.1)

Recall_D = TP_D / (TP_D + FN_D). (150.2)

This measures something deeper than ordinary predictive accuracy:

> does the agent know when its world is wrong?

---

151. Residual Attribution Accuracy

If ground truth is available, define:

A_R = fraction of anomalies correctly attributed to state-level versus declaration-level failure. (151.1)

The residual-ledger architecture predicts higher:

A_R

than an ordinary error-minimizing baseline.

This would directly test the claimed value of residual structure.

---

152. Recovery Without Identity Collapse

Performance after world revision is not enough.

A system might recover by discarding its entire past.

Therefore define continuity:

C_id = similarity of preserved purpose, valid history, and stable competencies across revision. (152.1)

The benchmark should reward:

high post-shift performance

and

high C_id.

The desired outcome is:

revision without amnesia. (152.2)

---

153. The Deepest Test of Retention

A difficult benchmark should deliberately reintroduce a structure that was previously rejected.

If L^- preserved it properly, the system should recover faster the second time.

Let:

T_1 = discovery latency during first encounter.

T_2 = rediscovery latency after the structure becomes relevant again.

The theory predicts:

$T_2 < T_1$. (153.1)

even when the rejected structure never entered the admitted ledger.

That would be a particularly strong demonstration of residual memory.

---

154. The Deepest Test of Closure

Present several locally successful models that cannot all coexist coherently.

Activation alone will show that each works in some context.

Closure must discover:

which relations can be integrated,

which require contextual separation,

and which remain residual.

This distinguishes:

local performance

from

world constitutability.

A system without a genuine Closure function should struggle here.

---

155. The Deepest Test of Selection

Construct a world-model containing redundant but performance-preserving structure.

A purely performance-driven learner may retain it.

Selection should compress it while preserving:

task success,

identity,

and future recoverability.

Thus Selection is tested not by immediate accuracy gain but by:

structural economy without destructive loss.

---

156. The Deepest Test of Generation

Create an ontology shift that cannot be solved by recombining current variables.

The system must introduce a genuinely new latent distinction.

If its candidate generator merely searches parameter space inside the old declaration, it fails.

Thus true Generation must occasionally enlarge or re-factor the representational language itself.

---

157. The Deepest Test of Activation

Provide many plausible hypotheses but limited experimentation budget.

The system must choose which candidates deserve causal exposure.

Thus Activation is not merely “turn everything on.”

It is:

resource-bounded enactment of possibility.

This makes A a decision problem distinct from G.

---

158. A Scientific Programme Emerges

The theory can therefore be developed through four increasingly demanding stages.

Stage I — Functional ablation

Do the five functions matter?

Stage II — Regime discovery

Do they form metastable modes?

Stage III — Topology discovery

Does productive and regulatory coupling resemble the proposed double cycle?

Stage IV — Cross-domain replication

Does the architecture recur beyond one implementation?

Only after Stage IV should strong claims of generality be entertained.

---

159. Cross-Domain Candidates

Potential test domains include:

- continual-learning agents;
- scientific-discovery systems;
- adaptive robotics;
- autonomous software agents;
- organizational decision systems;
- ecological adaptation models;
- human problem-solving protocols.

The same labels need not appear.

The question is whether the same **functional decomposition and transition logic** recur.

---

160. The Main Theoretical Contribution

The deepest claim of the article is therefore not:

“there are five stages.”

It is:

> **World revision requires a governance architecture that separates producing possibilities, making them consequential, constituting a temporary world, criticizing that world, and preserving enough history and unresolved remainder to make future revision possible.**

Five is the present minimal candidate for separating those jobs cleanly.

The double-cycle topology is a stronger secondary conjecture.

---

161. Conclusion

Most theories of adaptive intelligence ask how a system should update within a world.

Persistent intelligence faces a harder problem.

Its world can become wrong.

When that happens, neither prediction-error minimization nor unrestricted self-modification is sufficient.

The system must know:

what to preserve,

what to question,

what to enact,

what to commit to,

what to reject,

and when accumulated failure has become evidence against the world that produced the failure.

This article has proposed a candidate answer.

A persistent self-revising bounded system may require five functionally distinct control regimes:

Generation — producing viable alternatives.

Activation — exposing selected possibilities to consequence.

Closure — integrating a temporarily coherent and accountable world.

Selection — discriminating, pruning, and refining unsupported structure.

Retention — preserving successful history, unresolved remainder, and recovery capacity.

Their candidate productive circulation is:

$$G \rightarrow A \rightarrow C \rightarrow S \rightarrow R \rightarrow G. \quad (161.1)$$

A second independently motivated regulatory topology is:

$$G \dashv C \dashv R \dashv A \dashv S \dashv G. \quad (161.2)$$

The first enables constructive succession.

The second prevents each constructive function from becoming pathological through overexpression.

Both operate above a deeper SMFT mechanism:

world-relative dynamics,

observer-dependent trace formation,

dual trace–residual ledgering,

residual-driven revision pressure,

hysteretic latching,

and multi-depth self-revision.

The resulting system is neither a fixed pipeline nor a simple oscillator.

It is a history-bearing recursive circulation:

$S_{n+1} = \Phi(S_n, L^+_n, L^-_n)$. (161.3)

Every cycle changes the conditions of the next.

The proposed number five remains falsifiable.

Four may prove sufficient.

Six may prove necessary.

Different systems may support different coarse-grainings.

What matters scientifically is that the regime number, the transition topology, and the value of residual preservation can all be tested independently.

Only after this engineering architecture has been established do we encounter a striking historical comparison.

The independently derived mapping:

Generation ↔ Wood

Activation ↔ Fire

Closure ↔ Earth

Selection ↔ Metal

Retention ↔ Water

reproduces not only the traditional Wuxing generative sequence but, under the proposed cross-regulatory relationships, also its controlling sequence.

That correspondence cannot validate the modern theory.

Nor can the modern theory validate the traditional system.

But the double match is sufficiently structured to deserve investigation rather than dismissal as a five-label coincidence.

The more consequential possibility is that both are representations—at radically different historical and conceptual resolutions—of a recurring systems problem:

> **how a bounded world generates novelty, turns novelty into consequence, closes around a workable order, cuts away what cannot survive,**

preserves what must endure, and uses its accumulated remainder to begin again.

If that problem can be made experimentally precise, the old language of phases becomes optional.

The engineering problem remains.

And that engineering problem may be central to building systems that can do something much harder than learn:

revise the worlds within which their learning makes sense.

Yes. The **main article is essentially complete**, but I would add one final response containing the technical appendices and the explicit epistemic-status table. Those additions are important because they make it harder for readers to mistake the five-regime proposal for a numerological assertion.

Below is the continuation I would attach to the article.

Appendix A — Epistemic Status of the Claims

Because the framework combines established control concepts, explicit constructions, SMFT-derived concepts, and new conjectures, each major claim should be assigned an epistemic status.

Use the following notation:

[K] Known structure or standard mathematical fact

[D] Derived within the stated assumptions

[C] Explicit construction

[H] Testable hypothesis

[A] Structural analogy

[F] Falsified or superseded claim

This follows the methodological discipline already developed in the later SMFT audit, which explicitly separates known results, derived consequences, constructions, hypotheses, analogies, and superseded claims. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

The principal claims of this article can then be classified as follows.

Claim	Status
A bounded adaptive system can distinguish ordinary state update from world-model revision	[C]/[H] as a general architecture
Trace and residual can be stored in separate ledgers	[C]
Persistent directional residual can be used as a revision signal	[H]
Generation, Activation, Closure, Selection, and Retention are functionally distinguishable	[H1]
All five functions are necessary for robust persistent self-revision	[H1]
The five functions form five metastable control regimes	[H2]
Five is the minimal useful coarse-graining	[H3]

Claim	Status
G→A→C→S→R→G is a preferred productive circulation	[H4]
G⊣C⊣R⊣A⊣S⊣G is a stabilizing regulatory circulation	[H5]
The five-regime network has a 2+1+2 organization	[H6]
For five nodes, neighbour and non-neighbour five-cycles partition K_5	[K]
The resulting two-cycle graph corresponds to Wuxing Sheng/Ke topology	[A], conditional on the proposed mapping
Wuxing therefore scientifically validates the five-regime model	**Not claimed**
Five contemporary AI research traditions must map one-to-one onto the five regimes	**Not claimed in this article**

This table is important.

The article becomes significantly stronger once the reader can see exactly where formal structure ends and conjecture begins.

Appendix B — Minimal Formal Specification

A minimal Boundary Circulation Controller can be specified without committing to any particular neural architecture.

Let the full state be:

$$S = (D,P,x,L^{+},L^{-},a). \quad (B.1)$$

where:

D = current declaration

P = persistent reference or purpose

x = operational state

L^{+} = admitted trace ledger

L^{-} = residual ledger

a = regime activation vector.

Let:

$$a = (aG,aA,aC,aS,aR), \quad (B.2)$$

with:

$$a_i \geq 0, \quad \Sigma_i a_i = 1. \quad (B.3)$$

The operational state evolves as:

$$\dot{x} = \Sigma_i a_i f_i(x;D,P,L^{+},L^{-}). \quad (B.4)$$

Observation produces:

$$(T,r) = \Pi D,P,O,a(x). \quad (B.5)$$

The ledgers update according to:

$L^{+}_{t+1} = L^{+}_{t} \oplus T_t$. (B.6)

$L^{-}_{t+1} = L^{-}_{t} \oplus r_t$. (B.7)

Residual magnitude is:

$\rho_t = \|r_t\|$. (B.8)

Directional residual coherence over window W is:

$\delta R = \|\Sigma_{k \in W} w_k r_k\| / \Sigma_{k \in W} w_k \|r_k\|$. (B.9)

Revision pressure may be represented generically as:

$\Pi rev = \mathscr{R}(D,P,L^{+},L^{-},\rho,\delta R)$. (B.10)

The controller evolves according to:

$\dot{a} = Fcontrol(a,S,\Pi rev)$. (B.11)

Declaration revision occurs only when:

$\Pi rev > \Theta up$. (B.12)

A new declaration becomes latched when:

$\Pi rev < \Theta down, \quad \Theta down < \Theta up$. (B.13)

The hysteresis gap is:

$H = \Theta up - \Theta down$. (B.14)

Thus:

$H > 0$ (B.15)

suppresses frame chatter.

Purpose revision occurs on a slower timescale:

$\dot{P} = \varepsilon P \, UP(P,L^{+},L^{-}), \quad \varepsilon P \ll 1$. (B.16)

This is sufficient to specify the architecture abstractly.

Appendix C — Signed Five-Regime Interaction Model

Let the productive cycle be:

$G \rightarrow A \rightarrow C \rightarrow S \rightarrow R \rightarrow G$. (C.1)

Let the regulatory cycle be:

$G \dashv C \dashv R \dashv A \dashv S \dashv G$. (C.2)

For ordering:

(G,A,C,S,R),

define a signed interaction matrix W.

A schematic version is:

	G	A	C	S	R
G	0	+	−	0	0
A	0	0	+	−	0
C	0	0	0	+	−
S	−	0	0	0	+
R	+	−	0	0	0

(C.3)

The activation dynamics can be written:

$\dot{a}_i = a_i[b_i + \Sigma_j W_{ij} a_j - \varphi_i(a_i)]$. (C.4)

Here:

b_i = context-dependent regime drive

W_{ij} = cross-regime coupling

φ_i = saturation or self-limiting term.

Residual may deform the intrinsic drives:

$b_i = b_i^0 + \beta_i(L^-, P, D)$. (C.5)

In particular:

$\partial bG/\partial \delta R > 0$ (C.6)

is a natural candidate condition: persistent structured residual should increase pressure toward renewed Generation.

However, this should be tested rather than assumed universally.

Appendix D — Why the Five-Node Graph Is Mathematically Interesting

For five vertices, the complete undirected graph K_5 contains:

$|E(K_5)| = 5 \times 4/2 = 10$. (D.1)

A five-cycle contains five edges:

$|E(C_5)| = 5$. (D.2)

The complement of a five-cycle inside K_5 is itself another five-cycle:

$K_5 \setminus C_5 \cong C_5$. (D.3)

Therefore:

$E(K_5) = E(C_5) \cup E(C_5^*)$, (D.4)

with:

$E(C_5) \cap E(C_5^*) = \emptyset$. (D.5)

The proposed architecture assigns:

outer cycle → productive relations

inner cycle → regulatory relations.

Consequently every pair of regimes receives exactly one coarse relation type.

This property is mathematically exact **once five nodes have already been chosen**.

It does not derive the number five.

That logical direction must be preserved.

Nevertheless, it creates an unusually economical interaction grammar:

10 pairwise relationships

compressed into

2 cyclic rules.

This motivates an interaction-compression measure.

Let:

$N_{pair}(K) = K(K-1)/2$. (D.6)

Let R_{rule} denote the number of independent relational rules.

Define:

$C_{interaction} = N_{explained} / R_{rule}$. (D.7)

One empirical question is whether BCC-5 yields unusually high explanatory compression relative to BCC-4 and BCC-6 while retaining comparable predictive performance.

Appendix E — Competing Minimal Models

A serious experimental programme should begin with several candidate architectures.

E.1 Three-Regime Model

The simplest plausible decomposition is:

Explore → Commit → Conserve. (E.1)

This captures expansion, closure, and persistence.

Its likely weakness is that it conflates:

generation with enactment,

and:

selection with retention.

E.2 Four-Regime Model

The strongest four-regime null model appears to be:

G → A → CS → R. (E.2)

where:

CS = Closure + Selection.

Its major predicted failure is Constitution–Judgment Collapse.

The same subsystem determines:

what constitutes the frame

and

what evidence validates that frame.

The five-regime hypothesis predicts that separating C and S should improve robustness under frame-challenging evidence.

E.3 Five-Regime Model

The principal hypothesis is:

G → A → C → S → R. (E.3)

This is BCC-5.

E.4 Six-Regime Model

A plausible six-regime architecture splits Selection:

G → A → C → Evaluate → Refine → R. (E.4)

Another splits Retention:

G → A → C → S → Consolidate → Reactivate. (E.5)

The six-regime model should win if one of these subdivisions supports an independently metastable control state whose explicit separation significantly improves prediction or intervention.

Appendix F — Model Selection Protocol

The regime number must be inferred rather than celebrated.

For each candidate K, fit model MK.

Evaluate:

1. held-out predictive likelihood;
2. intervention accuracy;
3. regime dwell-time structure;
4. transition reproducibility;
5. ontology-shift recovery;
6. memory retention;
7. unnecessary declaration revisions;
8. computational complexity.

Define:

$$Q(K) = Fit(K) - \lambda_1 Complexity(K) - \lambda_2 InterventionError(K) - \lambda_3 RevisionCost(K). \quad (F.1)$$

Then:

$$K^* = \operatorname{argmax}_K Q(K). \quad (F.2)$$

The Five-Regime Minimality Hypothesis predicts:

$$K^* = 5 \quad (F.3)$$

for a nontrivial class of PSRBS environments.

Failure to obtain $K^*=5$ is not a failure of the experimental protocol.

It is precisely what makes the claim falsifiable.

Appendix G — Blind Regime Discovery Protocol

The most convincing experiment would withhold all five regime labels.

Step 1 — Generate environments

Construct environments containing both:

parameter changes

and:

ontology changes.

The agent does not receive labels distinguishing them.

Step 2 — Allow multiple repair depths

Permit:

state adaptation,

control-policy change,

declaration revision.

Step 3 — Preserve residual

Maintain L^- independently from admitted history.

Step 4 — Learn latent control modes

Infer:

$z_t \in \{1,\ldots,K\}$.

Do not preassign meanings.

Step 5 — Infer K

Compare:

K = 3,4,5,6,7,…

using held-out model selection.

Step 6 — Interventionally characterize modes

Perturb:

candidate generation,

action gain,

commitment threshold,

pruning pressure,

memory consolidation.

Determine whether distinct latent modes respond selectively.

Step 7 — Interpret only afterward

Only after mode discovery ask whether modes correspond to:

Generation,

Activation,

Closure,

Selection,

Retention.

This protocol minimizes retrospective fitting.

Appendix H — Predicted Ablation Failures

H.1 Remove Generation

Expected:

- failure to invent new abstractions;
- excessive parameter fitting;
- inability to escape obsolete declarations.

Signature:

high local adaptation, poor ontology-shift recovery.

H.2 Remove Activation

Expected:

- many hypotheses;
- insufficient causal testing;
- weak conversion from representation to evidence.

Signature:

high candidate diversity, low information gain from intervention.

H.3 Remove Closure

Expected:

- locally useful models remain fragmented;
- incompatible models coexist without explicit contextual separation;
- weak persistent world structure.

Signature:

good local performance, low cross-context coherence.

H.4 Remove Selection

Expected:

- redundant structures accumulate;
- contradictions persist;
- representational complexity grows.

Signature:

increasing K with little performance gain.

H.5 Remove Retention

Expected:

- rapid immediate adaptation;
- weak consolidation;
- catastrophic forgetting;
- poor recovery of previously rejected possibilities.

Signature:

high short-term plasticity, poor long-term continuity.

Appendix I — Predicted Edge-Ablation Failures

Removing a regime is crude.

A stronger test removes individual regulatory couplings.

Remove S ⊣ G

Prediction:

uncontrolled candidate proliferation.

Remove R ⊣ A

Prediction:

long activation episodes and slow recovery.

Remove G ⊣ C

Prediction:

excessive declaration persistence under world change.

Remove C ⊣ R

Prediction:

unbounded accumulation of latent alternatives and residual history.

Remove A ⊣ S

Prediction:

premature pruning and insufficient empirical testing.

The existence of these differentiated edge failures would provide stronger evidence for the dual-cycle topology than simply observing five nodes.

Appendix J — The Historical Wuxing Correspondence

Only after deriving the modern architecture should the comparison with **Wuxing (**五行**)** be introduced.

For an English-language audience, “Five Elements” can be misleading.

The relevant interpretation is closer to:

Five Phases

or:

Five Processes.

The proposed correspondence is:

Modern functional regime	**Wuxing comparison**
Generation / Differentiation	Wood
Activation / Amplification	Fire
Closure / Integration	Earth
Selection / Refinement	Metal
Retention / Latency	Water

Under this mapping:

$$G \rightarrow A \rightarrow C \rightarrow S \rightarrow R \rightarrow G \quad (J.1)$$

has the same abstract cyclic ordering as the traditional generative sequence.

More unusually:

$$G \dashv C \dashv R \dashv A \dashv S \dashv G \quad (J.2)$$

has the same abstract topology as the traditional controlling sequence.

The correct conclusion is narrow:

> The proposed engineering network and the traditional Wuxing graph possess a conditional structural correspondence under the stated mapping.

It does **not** establish historical equivalence, causal influence, or scientific validation.

Appendix K — Why the Historical Comparison Is Still Worth Keeping

Scientific caution does not require deleting every historical analogy.

A useful analogy can serve at least four legitimate purposes.

K.1 Compression

A complicated functional architecture becomes easier to remember.

K.2 Communication

The dual-cycle structure becomes immediately visually recognizable.

K.3 Hypothesis generation

Traditional descriptions may suggest candidate regulatory failures worth testing, provided those hypotheses are tested independently.

K.4 Cross-domain recognition

Researchers working in apparently unrelated domains may recognize the same functional pattern.

The danger begins only when:

analogy → evidence.

The correct direction is:

independent theory → structural comparison → new hypothesis → independent test.

Appendix L — The Future Article on Research Traditions

The next paper should begin only after this one is complete.

Its question should not be:

> Which five AI schools correspond to the Five Phases?

That framing is too weak.

The better question is:

> **Which contemporary research programmes have developed the strongest mathematics, engineering tools, or empirical methods for different parts of the independently derived Boundary Circulation architecture?**

Then one can examine whether particular programmes are especially concentrated around:

Generation,

Activation,

Closure,

Selection,

Retention,

or instead span several.

The correspondence should be many-to-many:

Research programme → weighted functional profile. (L.1)

For programme p define:

$w(p) = (wG,wA,wC,wS,wR)$. (L.2)

with:

$\Sigma_i w_i = 1.$ (L.3)

This is already more rigorous than assigning every school one fixed label.

A programme may have:

$wC = 0.55$,

$wS = 0.20$,

$wG = 0.15$,

and smaller weights elsewhere.

The visually dominant regime can still supply the iconography for an infographic without pretending that the programme exclusively "belongs" to that phase.

That will be the right mathematical basis for the later **Five Research Traditions / Five Dominant Functions** article.

Appendix M — The Intrinsic Mathematics of Five-Regime Boundary Circulation

M.1 Conditional Mathematical Universality

The main body of this article proposes an empirical hypothesis: a persistent self-revising bounded system may be usefully coarse-grained into five functionally distinct control regimes—Generation, Activation, Closure, Selection, and Retention.

That empirical claim may eventually prove correct, approximately correct, scale-dependent, or wrong.

The purpose of this appendix is different.

It asks whether the five-regime architecture, once abstracted away from all interpretation, possesses a mathematical structure that remains true independently of whether the model describes artificial intelligence, cognition, organizations, biology, or any physical system at all.

The answer is yes, provided the assumptions are stated precisely.

The resulting universality is not empirical universality. It is **structural universality**:

> A mathematical consequence is structurally universal when it depends only on an abstract relational architecture and not on the semantic or physical interpretation assigned to its elements.

Accordingly, the strongest claim of this appendix is not:

> Reality must contain five regimes.

It is:

> **If a finite system satisfies a particular dual-cycle relational grammar, then the number five and the associated five-cycle algebra follow mathematically rather than metaphorically.**

This distinction is essential.

M.2 Remove All Semantic Interpretation

Forget for the moment that the five states were called:

Generation,

Activation,

Closure,

Selection,

Retention.

Let there instead be an unknown number N of abstract states:

$Q = \{q_0, q_1, \ldots, q_{n-1}\}$. (M.1)

Suppose pairwise relationships among these states belong to two functional classes.

Call the first class:

productive relations.

Call the second:

regulatory relations.

The terminology is not important for the mathematics. They could equally be called relation A and relation B.

We now impose three axioms.

M.2.1 Axiom 1 — Productive Circulation

The productive relations form one simple cycle through all N states:

$G_+ \cong C_n$. (M.2)

Every state therefore has exactly two productive neighbours:

$\deg_+(q_i) = 2$. (M.3)

This captures the simplest possible homogeneous circulation.

M.2.2 Axiom 2 — Dual Relational Completeness

Every unordered pair of distinct states belongs to exactly one of the two relation classes.

Thus:

$E_+ \cap E_- = \emptyset$, (M.4)

and:

$E_+ \cup E_- = E(K_n)$. (M.5)

No pair is left unclassified.

No pair simultaneously belongs to both relation classes.

The two relation types jointly exhaust the complete pairwise relational structure.

M.2.3 Axiom 3 — Regulatory Circulation

The regulatory relations are structurally of the same minimal type as the productive relations:

$G_- \cong C_n$. (M.6)

Thus each state also has exactly two regulatory neighbours:

$\deg_-(q_i) = 2$. (M.7)

The question is now entirely mathematical:

> For which N can these three axioms simultaneously hold?

The answer is uniquely:

$N = 5$.

M.3 Dual-Cycle Completeness Theorem

Theorem M.1

If the complete pairwise relation set among N states can be partitioned into exactly two edge-disjoint Hamiltonian cycles, then $N = 5$.

Proof

A Hamiltonian cycle C_n contains exactly N edges:

$|E(C_n)| = N$. (M.8)

Two edge-disjoint Hamiltonian cycles therefore contain:

$|E_+| + |E_-| = 2N$. (M.9)

The complete graph K_n contains:

$|E(K_n)| = N(N-1)/2$. (M.10)

Dual relational completeness requires:

$2N = N(N-1)/2$. (M.11)

For $N > 0$:

$4 = N-1.$ (M.12)

Therefore:

$N = 5.$ (M.13)

QED.

This is already enough to establish the central result:

> **Exactly two equal-status cyclic relation classes can exhaust all pairwise relations only for five states.**

No appeal to SMFT, AI, Wuxing, cognition, or physical interpretation occurs anywhere in the proof.

M.4 The Same Result From Local Degree Counting

The theorem has an even simpler local form.

Every state possesses:

two productive neighbours,

and:

two regulatory neighbours.

If these relations exhaust all other states, then:

$2 + 2 = N-1.$ (M.14)

Hence:

$N = 5.$ (M.15)

From the viewpoint of every state:

2 productive neighbours

- 1 self
- 2 regulatory neighbours
  = 5 total states.

Thus:

$2 + 1 + 2 = 5.$ (M.16)

This is worth distinguishing from the earlier functional 2+1+2 interpretation of the five-regime architecture.

The earlier interpretation concerned:

two outward functions,

one closure function,

two inward functions.

Equation (M.16) is different.

It is an intrinsic relational property of the five-cycle:

two adjacent states,

the reference state itself,

two nonadjacent states.

The two 2+1+2 structures may later prove conceptually related, but one should not infer the functional interpretation from the graph identity alone.

M.5 The Complement Theorem

There is a third equivalent way to see the same result.

In C_n every vertex has degree:

$$\deg(C_n) = 2. \quad (M.17)$$

A vertex of K_n has degree:

$$\deg(K_n) = N-1. \quad (M.18)$$

Therefore, in the complement of the cycle:

$$\deg(\bar{C}_n) = N-3. \quad (M.19)$$

If the complement is itself to be another cycle, then:

$$\deg(\bar{C}_n) = 2. \quad (M.20)$$

Therefore:

$$N-3 = 2. \quad (M.21)$$

and again:

$$N = 5. \quad (M.22)$$

Thus:

$$\bar{C}_5 \cong C_5. \quad (M.23)$$

This gives an exact graph-theoretic characterization:

C_5 is the unique cycle graph whose complement is itself another cycle.

This is the mathematical reason a pentagonal architecture can support two structurally equivalent relation classes that together account for every pair.

M.6 Why Four Is Structurally Different

For four states:

$G_{+} \cong C_4$. (M.24)

Each vertex has two productive neighbours.

Only one nonproductive neighbour remains.

Hence:

$\deg(C_4) = 1$. (M.25)

The complement is:

$\bar{C}_4 \cong 2K_2$. (M.26)

That is, two disconnected pairs.

A four-state system can therefore possess:

one four-cycle

plus:

one pairing relation.

What it cannot possess is two equal-status circulation graphs that jointly exhaust all pairwise interactions.

This does not make four-state architectures inferior in general.

It means that four does not solve the **dual-cycle completeness problem**.

M.7 Why Six Is Structurally Different

For six states:

$\deg(C_6) = 3$. (M.27)

Each state has:

two productive neighbours

but:

three complementary neighbours.

The second relation class is therefore more connected than the first.

The complement of C_6 is not C_6.

A six-state architecture can certainly possess rich mathematics, but the simple symmetry:

two productive relations

versus

two regulatory relations

has disappeared.

Thus five is not special because “five is a good number.”

It is special under the much narrower condition:

> two homogeneous cyclic relation classes should completely classify every pair.

M.8 The Cyclic Group Representation

Once N = 5, the states can be labeled naturally by the cyclic group:

$\mathbb{Z}_5 = \{0,1,2,3,4\}$. (M.28)

Let one-step productive circulation be generated by:

$P(i) = i+1 \bmod 5$. (M.29)

Then:

$P^5 = I$. (M.30)

The productive cycle is therefore:

$0 \to 1 \to 2 \to 3 \to 4 \to 0$. (M.31)

Its undirected relation class is:

$\pm 1 \bmod 5$. (M.32)

What pairwise relationships remain?

Only:

$\pm 2 \bmod 5$. (M.33)

These are precisely the non-neighbour relations.

Thus the complete set of nonidentity relative positions splits into:

$\mathbb{Z}_5 \setminus \{0\} = \{\pm 1\} \cup \{\pm 2\}$. (M.34)

This partition is exhaustive and disjoint.

That is the algebraic core of the double-cycle architecture.

M.9 The Regulatory Cycle as a Power of the Same Generator

Choose one orientation for the complementary relation:

$K(i) = i+2 \bmod 5$. (M.35)

Then:

$K = P^2$. (M.36)

Because 2 and 5 are coprime, repeatedly applying P^2 traverses all five states:

$0 \to 2 \to 4 \to 1 \to 3 \to 0$. (M.37)

Hence:

$K^5 = I$. (M.38)

and K itself generates another five-cycle.

This has an important implication.

The productive and regulatory cycles do not require two unrelated primitive operators.

Both arise as powers of one cyclic generator:

P.

The nonidentity powers are:

P^1,

P^2,

P^3,

P^4. (M.39)

These exhaust all directed relative positions.

They can be interpreted abstractly as:

P^1 = forward relation of class 1,

P^4 = inverse relation of class 1,

P^2 = forward relation of class 2,

P^3 = inverse relation of class 2. (M.40)

Thus the entire directed pairwise relation algebra is generated by one operator satisfying:

$P^5 = I$. (M.41)

M.10 Uniqueness Up to Orientation

Once the productive C_5 has been fixed, its complement is fixed as an undirected graph.

Only orientation remains free.

The complementary cycle may be directed as:

$i \to i+2 \bmod 5$, (M.42)

or:

$i \to i-2 \bmod 5.$ (M.43)

The two possibilities are reversals of one another.

Therefore the regulatory topology is unique up to:

rotation,

reflection,

and direction reversal.

This is a substantial rigidity result.

If a five-regime empirical system really exhibits the dual-cycle completeness axioms, the second cycle is not one arbitrary choice among many graphs.

At the undirected level it is forced.

M.11 The Complete Pairwise Structure Is K_5

Five states generate:

$5 \times 4/2 = 10$ (M.44)

unordered pairs.

The productive cycle contributes five:

$|E_+| = 5.$ (M.45)

The regulatory cycle contributes the remaining five:

$|E_-| = 5.$ (M.46)

Hence:

$K_5 = C_5^{(+)} \oplus C_5^{(-)},$ (M.47)

where $\oplus$ denotes an edge-disjoint union.

This means that every possible pair of regimes belongs to exactly one relation class.

No extra edge type is required.

No edge remains unexplained.

No third relational category is needed.

This is one source of the unusual compression of the five-state architecture.

M.12 An Interaction-Compression Interpretation

Suppose a theory had to specify ten pairwise relations separately.

Its relational complexity would be high.

The dual-cycle architecture replaces those ten independent specifications with two rules:

Rule 1:

nearest cyclic neighbours belong to relation class +.

Rule 2:

non-nearest cyclic neighbours belong to relation class −.

Thus a small relational grammar generates the entire K_5 structure.

One can express this informally as:

10 pairwise relations ← 2 cyclic rules. (M.48)

A possible interaction-compression measure is:

C_int = N_explained / N_rules. (M.49)

For the idealized double-cycle architecture:

C_int = 10/2 = 5. (M.50)

This number itself should not be overinterpreted.

The important point is that the architecture provides **complete relational coverage with very low descriptive complexity**.

M.13 Generalization: Five Is Not the Only Beautiful Odd Cycle

To avoid a misleading impression, the five-state result should be placed inside a broader mathematical family.

For any odd complete graph K_{2m+1}, it is possible to decompose all edges into m edge-disjoint Hamiltonian cycles.

This is a classical Hamiltonian decomposition of odd complete graphs.

Thus:

$K_{2m+1} = C_{2m+1}^{(1)} \oplus C_{2m+1}^{(2)} \oplus \ldots \oplus C_{2m+1}^{(m)}$. (M.51)

The number of cyclic relation classes is:

m = (N−1)/2. (M.52)

Hence:

N = 2m+1. (M.53)

Examples are:

m=1 → N=3,

m=2 → N=5,

$m=3 \rightarrow N=7,$

$m=4 \rightarrow N=9.$ (M.54)

This generalization is important.

It shows that five is not universally privileged among all finite relational systems.

It is uniquely privileged for the case:

$m = 2.$

That is:

> **exactly two complete cyclic pairwise relation classes.**

M.14 Why Five Is the Minimal Dual-Relation Architecture

For three states:

$m = 1.$ (M.55)

Only one cyclic relation class exists.

There is no independent second pairwise class.

Therefore three is too small for a complete distinction between:

productive

and:

regulatory

cyclic relations.

For five states:

$m = 2.$ (M.56)

Two complete relation classes exist.

For seven states:

$m = 3.$ (M.57)

Three relation classes are required for complete pairwise coverage.

Thus five is exactly the smallest odd cyclic architecture that can support:

two distinct homogeneous relation classes

and no third class.

This gives five a genuine minimal property.

M.15 A General Relational Principle

The previous result can be written compactly.

If a symmetric cyclic architecture possesses m complete pairwise relation classes, then:

$N = 2m+1.$ (M.58)

Therefore:

one relation class → three states,

two relation classes → five states,

three relation classes → seven states,

and so forth.

Under the present theory, the important empirical assumption is:

$m = 2$,

corresponding to:

productive/enabling interaction,

and:

regulatory/restraining interaction.

If real systems require a third fundamental pairwise relation type, then one should not expect the dual-cycle theorem to select five.

This is an important limitation and an important falsifier.

M.16 The Mathematics Does Not Derive the Five Functions

This appendix must not reverse the logical direction of the article.

The graph theorem does **not** derive the semantic functions:

Generation,

Activation,

Closure,

Selection,

Retention.

Those functions emerged from the earlier engineering analysis of persistent self-revising systems.

Mathematics establishes something different.

It says:

> If those or any other states instantiate exactly two complete cyclic relation classes, the number of states must be five.

Conversely:

> If one independently establishes five states arranged in one productive cycle, and all remaining pairwise relations form one homogeneous second class, the second cycle is mathematically forced.

That is the correct scope.

M.17 Two Independent Routes Toward Five

This produces an important convergence.

Route A — Functional Minimality

The engineering analysis suggested five functions because merging them creates characteristic pathologies:

Generation ≠ Activation,

Activation ≠ Closure,

Closure ≠ Selection,

Selection ≠ Retention,

Retention ≠ Generation.

This produced:

N ≈ 5. (M.59)

Route B — Relational Completeness

The relational analysis asks for:

two edge-disjoint cyclic relation classes

that exhaust all pairs.

This produces exactly:

N = 5. (M.60)

The two arguments are logically independent.

One concerns functional differentiation.

The other concerns relational topology.

Their convergence is therefore more interesting than merely finding another way to draw a

pentagon.

M.18 The Third Convergence: Productive and Regulatory Semantics

The proposed five functional regimes also admit a natural interpretation of the two relation classes.

The productive cycle is:

$$G \to A \to C \to S \to R \to G. \quad (M.61)$$

The proposed regulatory cycle is:

$$G \dashv C \dashv R \dashv A \dashv S \dashv G. \quad (M.62)$$

Under the labeling:

$$G=0,$$

$$A=1,$$

$$C=2,$$

$$S=3,$$

$$R=4, \quad (M.63)$$

these become:

$$\text{productive: } i \to i+1 \bmod 5, \quad (M.64)$$

$$\text{regulatory: } i \to i+2 \bmod 5. \quad (M.65)$$

Thus the semantic proposal lands directly on the two algebraically available relation classes:

$$\pm 1$$

and:

$$\pm 2.$$

That is a nontrivial internal coherence result.

It does not validate the empirical model, but it shows that the proposed semantics fit the intrinsic relational geometry cleanly.

M.19 Circulant Linear Dynamics

The cyclic representation also produces a natural family of linear interaction operators.

Let P be the 5×5 cyclic shift matrix:

$$P^5 = I. \quad (M.66)$$

Any translation-invariant linear operator on $\mathbb{Z}_5$ has the circulant form:

$$W = c_0 I + c_1 P + c_2 P^2 + c_3 P^3 + c_4 P^4. \quad (M.67)$$

If reversal symmetry is imposed:

$$c_1 = c_4, \quad (M.68)$$

$$c_2 = c_3. \quad (M.69)$$

Then:

$$W = c_0 I + \alpha(P + P^{-1}) + \beta(P^2 + P^{-2}). \quad (M.70)$$

If nearest-neighbour relations are productive and non-neighbour relations regulatory:

$$W = c_0 I + \alpha(P + P^{-1}) - \beta(P^2 + P^{-2}), \quad \alpha, \beta > 0. \quad (M.71)$$

This gives the double-cycle architecture a natural dynamical operator.

M.20 Exact Fourier Decomposition

Circulant matrices are diagonalized by the discrete Fourier basis.

Let:

$$\omega = \exp(2\pi i/5). \quad (M.72)$$

Define:

$$v_k = (1, \omega^k, \omega^{2k}, \omega^{3k}, \omega^{4k})^T, \quad k = 0,1,2,3,4. \quad (M.73)$$

Then:

$$P v_k = \omega^k v_k. \quad (M.74)$$

For the symmetric signed operator in (M.71):

$$W v_k = \lambda_k v_k, \quad (M.75)$$

with:

$$\lambda_k = c_0 + 2\alpha \cos(2\pi k/5) - 2\beta \cos(4\pi k/5). \quad (M.76)$$

Thus the system can be decomposed exactly into five normal modes.

Questions about:

stability,

oscillation,

symmetry breaking,

mode amplification,

and regulatory gain

can therefore be studied analytically.

The five-regime structure is not merely a visual taxonomy. It supports an exact spectral calculus.

M.21 The Ordinary Five-Cycle Spectrum

For the unweighted adjacency matrix of C_5:

$\lambda_k = 2\cos(2\pi k/5).$ (M.77)

The resulting spectrum is:

$\{2, 1/\varphi, 1/\varphi, -\varphi, -\varphi\},$ (M.78)

where:

$\varphi = (1+\sqrt{5})/2.$ (M.79)

The golden ratio therefore appears automatically in the spectral geometry of C_5.

It also appears in the geometry of the regular pentagon and pentagram.

This is a standard mathematical property of fivefold cyclic geometry.

No physical, cognitive, or metaphysical significance follows from the occurrence of φ by itself.

That caveat is important.

The correct inference is:

fivefold cyclic symmetry generates golden-ratio quantities.

The reverse inference:

golden-ratio observation implies five-regime boundary dynamics

would not be justified.

M.22 Dihedral Symmetry

The cyclic group $\mathbb{Z}_5$ captures oriented circulation.

If reversal symmetry is included, introduce a reflection operator J satisfying:

$J^2 = I.$ (M.80)

Together with:

$P^5 = I,$ (M.81)

and:

$JPJ = P^{-1},$ (M.82)

the operators generate the dihedral group:

D_5. (M.83)

D_5 contains:

five rotations

and:

five reflections.

This provides the natural full symmetry group of the regular five-cycle.

In future work, reversal may acquire interpretations involving:

rollback,

counter-circulation,

reconstruction of previous declarations,

or inverse traversal.

None of those interpretations is required for the mathematics.

M.23 Approximate Structural Universality

Real systems are unlikely to exhibit exact C_5 graphs.

The useful empirical question is therefore whether their interaction networks approximate the ideal structure.

Let E_+ and E_- denote empirically inferred productive and regulatory edges.

Define overlap defect:

$\varepsilon_overlap = |E_+ \cap E_-| / |E(K_n)|$. (M.84)

Define missing-edge defect:

$\varepsilon_missing = |E(K_n) \setminus (E_+ \cup E_-)| / |E(K_n)|$. (M.85)

Let:

$\varepsilon_+ = d(G_+, C_n)$, (M.86)

$\varepsilon_- = d(G_-, C_n)$, (M.87)

where d is an appropriate graph distance.

Then define:

$\Delta_dual = w_1\varepsilon_overlap + w_2\varepsilon_missing + w_3\varepsilon_+ + w_4\varepsilon_-$. (M.88)

The ideal architecture satisfies:

$\Delta_dual = 0$. (M.89)

Empirical research can therefore ask whether:

Δ_{dual}

is systematically minimized near:

$N = 5.$ (M.90)

This converts the abstract theorem into a measurable structural hypothesis.

M.24 A New Experimental Strategy

The theorem suggests that the empirical search should not begin by asking:

“Can we identify five states?”

A stronger protocol is:

infer the states,

infer their pairwise couplings,

classify those couplings independently,

then ask whether two dominant relation classes emerge.

If the productive graph approximates one C_n and the regulatory graph approximates another C_n, while their union nearly exhausts K_n, the theorem predicts:

$N \approx 5.$

Thus the number five can become a **consequence of observed relational structure**, not an imposed prior.

That would be a much stronger empirical result.

M.25 What a Four-State Result Would Mean

Suppose blind experiments repeatedly infer four regimes.

If their productive topology remains approximately C_4, then the complementary graph cannot be another cycle.

One should instead expect something more like:

two opposing regulatory pairs.

Thus a four-state result would not simply say:

“five was wrong.”

It would predict a qualitatively different regulatory architecture.

That provides a sharper test.

M.26 What a Six-State Result Would Mean

If six states emerge and the productive topology is C_6, each node has three complementary relations.

A single simple regulatory cycle can account for only two of them.

Therefore at least one additional relational distinction must appear.

A six-state result may indicate that the current theory has compressed two different regulatory functions into one class.

This gives a principled way to interpret failure of the five-state model.

M.27 What a Seven-State Result Would Mean

Seven states naturally support three Hamiltonian relation classes in a complete decomposition.

Thus an empirical seven-state architecture could suggest:

three fundamental interaction families

rather than two.

For example, a future theory might discover:

productive,

regulatory,

and mediating

relations.

The present theory makes no such claim.

The point is that the mathematics already tells us what kind of conceptual enrichment a different regime number might require.

M.28 Structural Universality Versus Ontological Universality

The distinction can now be stated precisely.

Ontological universality

“All real systems possess five regimes.”

Not established.

Functional universality

“All persistent self-revising systems must implement exactly the five proposed functions.”

Not established.

Modal universality

“The five functions always form five metastable regimes.”

Not established.

Structural universality

“Every architecture satisfying Dual-Cycle Completeness has exactly five states and the C_5/C_5 relational structure.”

Established mathematically.

These claims must never be conflated.

M.29 The Mathematics Survives Failure of the Empirical Theory

Suppose future experiments show that:

four regimes outperform five,

or:

Closure and Selection are not separable,

or:

the regulatory topology is not cyclic,

or:

SMFT itself requires major revision.

The mathematical theorem remains unchanged.

If:

Axiom M1,

Axiom M2,

and:

Axiom M3

hold for any abstract system whatsoever, then:

$N = 5.$ (M.91)

That statement is independent of SMFT.

This is the sense in which the mathematical structure possesses truth “by itself.”

M.30 Why This Matters Philosophically

There are two very different ways for a theory to use mathematics.

The weaker way is to decorate a conceptual narrative with equations.

The stronger way is to discover that a conceptual architecture falls inside a rigid mathematical object whose consequences no longer depend on the original narrative.

The five-regime hypothesis has now reached at least a preliminary version of the second situation.

Its semantic interpretation remains conjectural.

But once the dual-cycle axioms are imposed, the graph structure is no longer negotiable.

Five follows.

The complementary cycle follows.

The $\mathbb{Z}_5$ relational algebra follows.

The circulant representation follows.

The Fourier decomposition follows.

These consequences are not matters of interpretation.

M.31 Three Levels of Closure in the Present Theory

The entire article can now be viewed as reaching three different kinds of closure.

Empirical closure remains open

Do real self-revising systems instantiate the architecture?

Unknown.

Functional closure is provisional

Are G, A, C, S, and R the correct minimal decomposition?

Testable hypothesis.

Mathematical closure is exact

Given Dual-Cycle Completeness:

$N = 5$.

This last result is complete within its premises.

That makes it a useful anchor for the otherwise exploratory theory.

M.32 A Stronger Final Interpretation of the Five-Regime Hypothesis

The five-regime proposal can now be stated more carefully.

It is not merely:

> We identified five useful functions.

It is:

> Functional analysis independently suggests five non-equivalent control roles. Relational analysis independently shows that exactly five states are required by a complete two-cycle interaction grammar. The proposed productive and regulatory relationships among the five functional roles then occupy precisely the two available relational classes of $\mathbb{Z}_5$.

This is a three-part coherence:

functional,

relational,

algebraic.

That is substantially stronger than the original five-school analogy from which the investigation began.

M.33 The Historical Wuxing Correspondence Revisited

Only now does the historical comparison acquire its proper epistemic position.

The modern reasoning has produced:

five functions,

a productive five-cycle,

a regulatory five-cycle.

Traditional Wuxing also presents:

five phases,

a Sheng (生, generative) cycle,

a Ke (克, controlling or restraining) cycle.

Under the tentative mapping:

Generation ↔ Wood,

Activation ↔ Fire,

Closure ↔ Earth,

Selection ↔ Metal,

Retention ↔ Water,

the two abstract directed topologies coincide.

That is a legitimate structural observation.

But the logical direction must remain:

modern functional derivation

→ mathematical analysis

→ historical comparison.

Never:

historical fivefold structure

→ proof of modern architecture.

M.34 What the Historical Match Adds

The Wuxing correspondence does not add mathematical validity.

The theorem is already valid without it.

What the historical correspondence adds is a new question:

> Why should a traditional qualitative process grammar and an independently developed modern boundary-control hypothesis land on the same unusually compact five-node dual-cycle topology?

Possible explanations include:

coincidence,

flexible reinterpretation,

historical accumulation of practical systems observations,

generic cognitive preference for fivefold schemes,

or:

a genuinely recurring control architecture.

Nothing in the present work can distinguish those explanations.

That question belongs to future comparative research.

M.35 A Compact Theorem Box for the Main Article

If the article needs a short highlighted statement, I would use:

Dual-Cycle Completeness Theorem.
Let N states possess two relation classes such that each class forms a Hamiltonian cycle, the two cycles are edge-disjoint, and together they classify every unordered pair of distinct states. Then N=5. Equivalently, C_5 is the unique cycle whose complement is also a cycle.

And immediately below:

Interpretive consequence.
If productive and regulatory interactions in a self-revising control architecture satisfy these conditions, five regimes are mathematically forced by the interaction grammar rather than inserted as a symbolic preference.

This is concise enough to become one of the memorable results of the paper.

M.36 A Compact Algebra Box

Likewise:

$Q \cong \mathbb{Z}_5$. (M.92)

$P(i)=i+1 \bmod 5$. (M.93)

$P^5=I$. (M.94)

Productive relation: ±1. (M.95)

Regulatory relation: ±2. (M.96)

$K=P^2$. (M.97)

$K^5=I$. (M.98)

$\mathbb{Z}_5\{0\}=\{\pm1\}\cup\{\pm2\}$. (M.99)

These equations summarize the relational algebra in almost minimal form.

M.37 A Compact Dynamical Box

For the symmetric signed controller:

$W = c_0 I + \alpha(P+P^{-1}) - \beta(P^2+P^{-2})$. (M.100)

The Fourier eigenvalues are:

$\lambda_k = c_0 + 2\alpha \cos(2\pi k/5) - 2\beta \cos(4\pi k/5)$. (M.101)

Thus stability of the homogeneous five-regime interaction model can be studied mode by mode.

This gives the conceptual theory a direct path toward simulation and analytical control theory.

M.38 Final Caution: Mathematical Beauty Is Not Empirical Evidence

This appendix uncovers an elegant structure.

That elegance must not be used as evidence that nature realizes it.

Many mathematically beautiful systems are never instantiated by the phenomenon one hoped they would describe.

The correct scientific sequence remains:

derive,

formalize,

predict,

intervene,

compare alternatives,

and revise.

The five-regime theory itself should obey the revision discipline it recommends to other systems.

If residual evidence accumulates against five, five must be relinquished.

M.39 Closing Remark — What Remains True Even If the Theory Fails

The Five-Regime Boundary Circulation Hypothesis may eventually prove correct, approximately correct, scale-dependent, or wrong.

That question belongs to experiment.

The mathematical statement established in this appendix is of a different kind.

If a finite relational architecture possesses exactly two edge-disjoint cyclic relation classes that jointly exhaust all pairwise relations, then it necessarily possesses five states.

Its productive cycle is C_5.

Its complementary relation graph is another C_5.

The two cycles partition K_5.

Their directed relations can be generated from a single order-five operator on $\mathbb{Z}_5$.

Homogeneous linear interactions on this structure are circulant and admit an exact Fourier decomposition.

None of these facts depends on what the five states represent.

They remain true if the states describe intelligent control regimes, ecological phases, organizational modes, abstract symbols, or nothing physical whatsoever.

This is the precise sense in which the structure possesses **mathematical universality independent of empirical interpretation**.

The mathematics therefore does not prove that intelligent worlds must be fivefold.

It establishes something narrower, but stronger:

> **if intelligent world-formation ever realizes a complete dual-cycle grammar of productive and regulatory relations, five is not a metaphor imposed upon the system afterward. Five is already contained in the grammar itself.**

And this changes the research question.

The question is no longer:

> Why should we believe in five?

It becomes:

> **Does persistent self-revising intelligence actually instantiate the relational axioms that make five inevitable?**

That is an empirical question.

The theorem is not.

Final Research Proposition

The entire present article can ultimately be compressed into one proposition:

> **Boundary Circulation Proposition.**
> Persistent self-revising bounded systems must solve several functionally conflicting tasks: they must generate alternative possibilities, enact selected possibilities, integrate a temporary working world, discriminate unsupported structure, and preserve both successful history and unresolved remainder. A five-regime architecture separating these functions is proposed as a minimal coarse-grained controller. Productive transitions and cross-regulatory restraints jointly create a history-bearing circulation whose switching is driven partly by structured residual. The number five, its metastability, and its proposed interaction topology are empirical hypotheses rather than premises.

And its strongest speculative extension is:

> **Dual-Cycle Conjecture.**
> If five such regimes are indeed the appropriate coarse-graining, a robust controller may organize their ten pairwise relationships into two complementary five-cycles: one enabling productive succession and one preventing pathological overexpression.

That is, I think, the right place to end **this** foundational article.

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are encouraged to approach the ideas with critical thinking and to consult primary scientific literature where appropriate.

This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.